

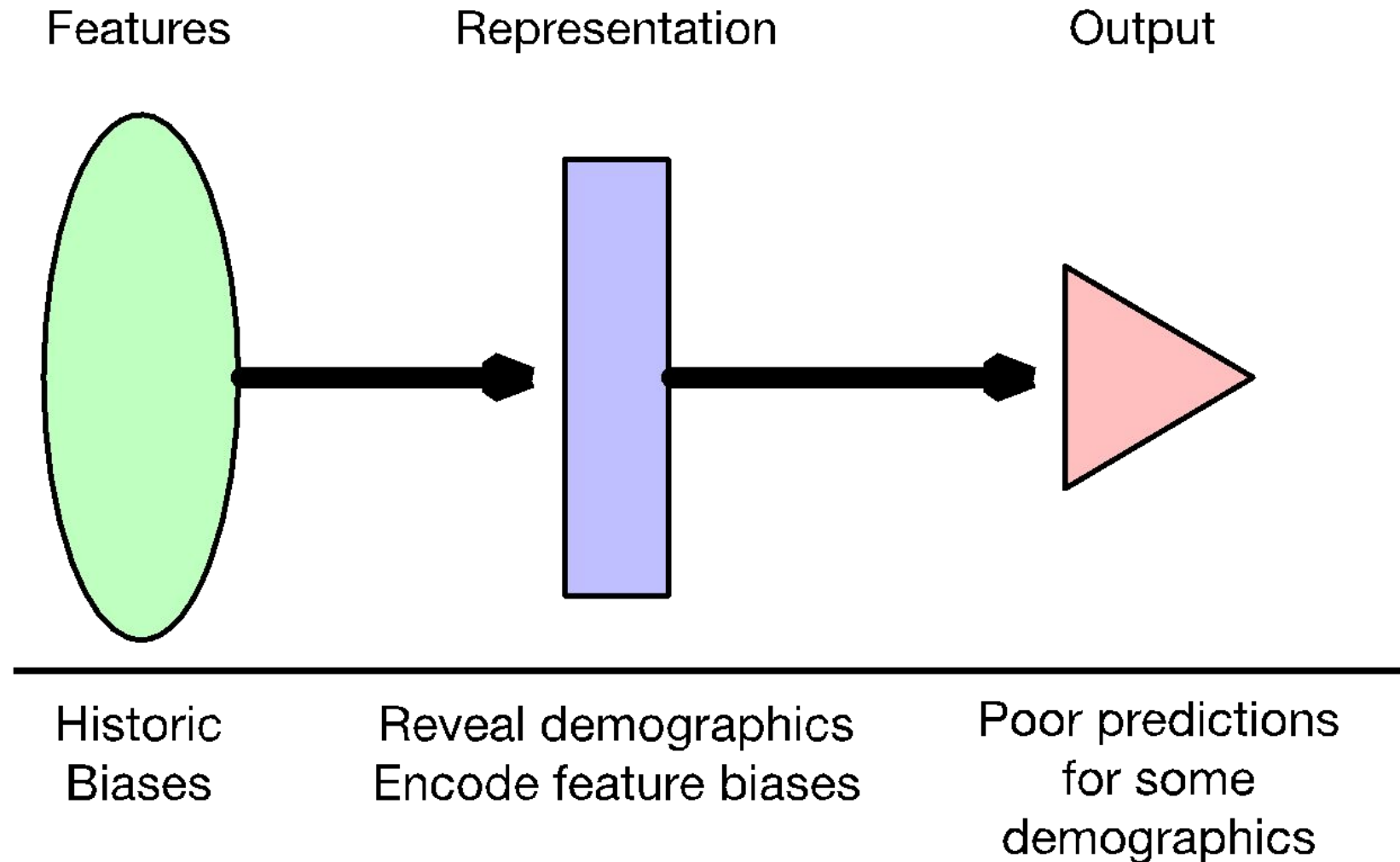
Evaluating Biases and Debiasing Methods

Fiona Guo

(Slides borrowed from Fred Morstatter)

April 16, 2025

Biases in Different Stages



Unfairness at different levels

Class	Representative examples
Discrimination via Input	<i>Employment:</i> Recruiting tool believes that men are more qualified and shows bias against women. <i>Loan Approval:</i> Loan eligibility system negatively rates people belonging to certain ZIP code, causing discrimination for certain races. <i>Criminal Justice:</i> Recidivism prediction system predicts black inmates are three times more likely to be classified as “high risk” than white inmates.
Discrimination via Representation	<i>Medical Image Diagnosis:</i> CNN model could identify patients’ self-reported sex from a retina image, and shows discrimination based on gender. <i>Credit Scoring:</i> Using raw texts as input, demographic information of authors is encoded in the intermediate representations DNN-based credit scoring classifiers. ⁵
Prediction Quality Disparity	<i>Facial Recognition:</i> Facial recognition performs very poorly for females with darker skin. <i>Language processing:</i> Language identification models perform significantly worse when processing text produced by people belonging to certain races. <i>Readmission:</i> ICU mortality and psychiatric 30-day readmission model prediction accuracy is significantly different across gender and insurance types.

Methods for Bias Mitigation

TABLE 2. Representative algorithms for mitigating unfairness in DNN models.

Class	Preprocessing	In-processing	Postprocessing
Discrimination via Input	Sensitive features removal	Attribution regularization ^{14,15}	Calibrated distribution ⁶
	Sensitive features replacement Reweighing ¹⁷ Optimized pre-processing ¹⁹	Reduction game ¹⁶ Prejudice remover ¹⁸	Calibrated equalized odds ⁹
Discrimination via Representation	Balanced dataset collection	Adversarial training ^{2,5} Adversarial fairness desideratum ²⁰ Semantic constraints ²¹ Distance metrics ^{22,23}	Troubling neurons turn OFF
Prediction Quality	Diverse dataset collection ²⁴	Transfer learning ²⁵	
Disparity	Synthetic data generation ²⁶	Multitask learning ²⁷	

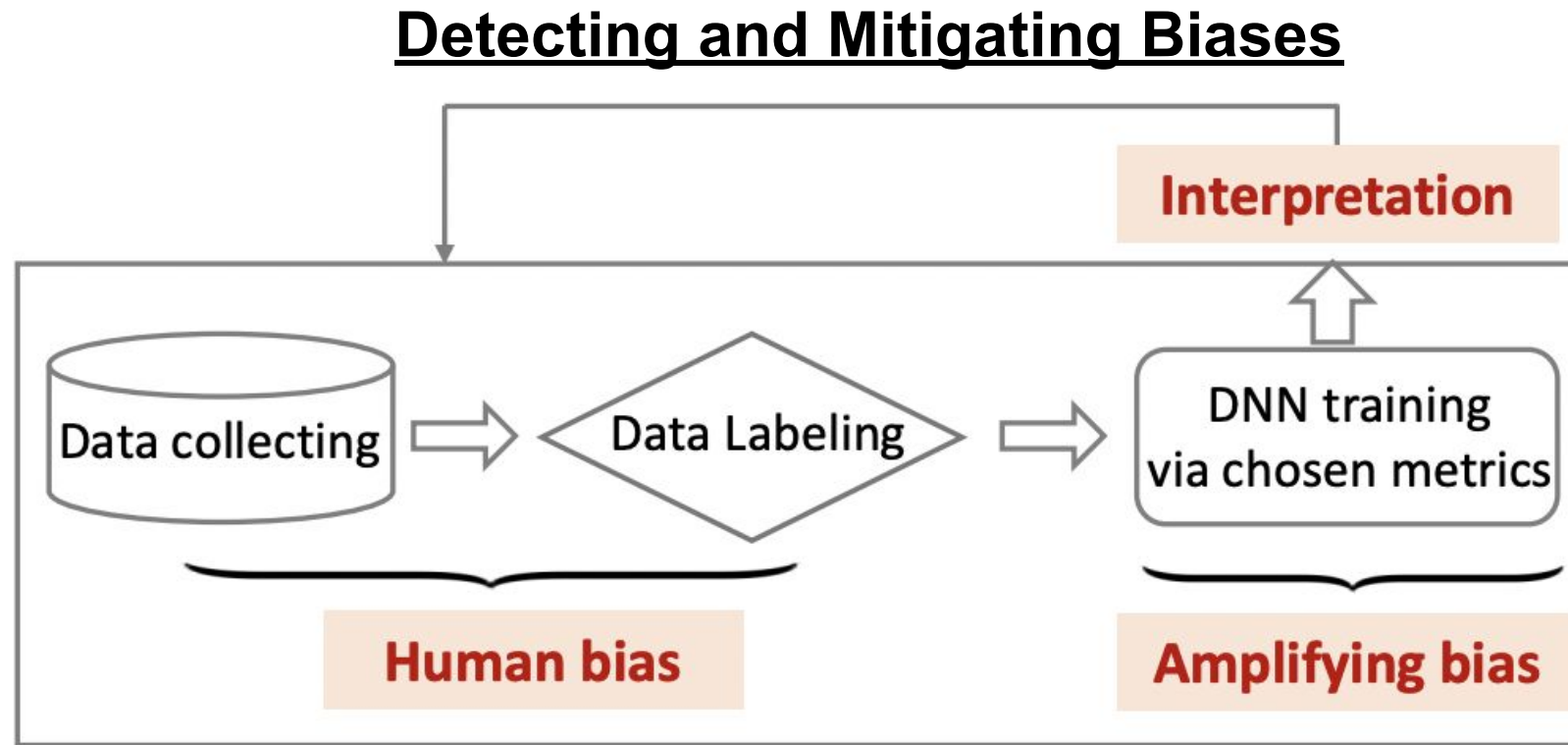
Preprocessing, in-processing, and postprocessing correspond to three stages of deep learning pipeline: dataset construction, model training, and model inference.

Fairness in Deep Learning: A Computational Perspective

Mengnan Du, Fan Yang, Na Zou, Xia Hu

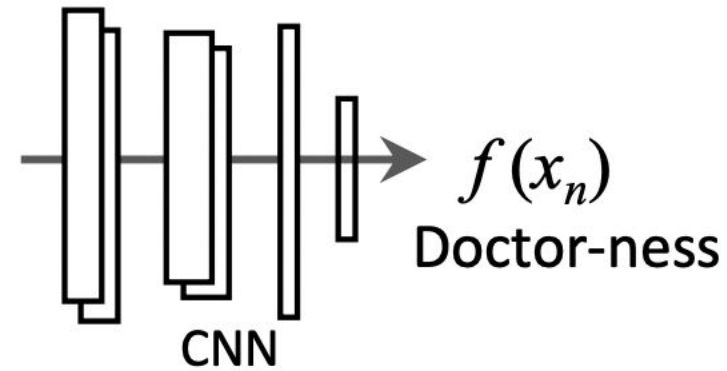
Abstract—Deep learning is increasingly being used in high-stake decision making applications that affect individual lives. However, deep learning models might exhibit algorithmic discrimination behaviors with respect to protected groups, potentially posing negative impacts on individuals and society. Therefore, fairness in deep learning has attracted tremendous attention recently. We provide a review covering recent progresses to tackle algorithmic fairness problems of deep learning from the computational perspective. Specifically, we show that interpretability can serve as a useful ingredient to diagnose the reasons that lead to algorithmic discrimination. We also discuss fairness mitigation approaches categorized according to three stages of deep learning life-cycle, aiming to push forward the area of fairness in deep learning and build genuinely fair and reliable deep learning systems.

Biases within a DNN



Biases

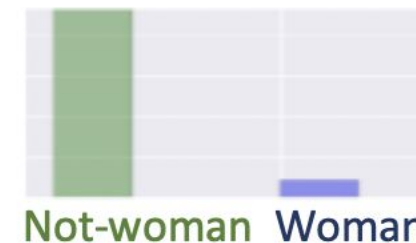
Biases within a DNN



Was **gender** concept important to this doctor image classifier?



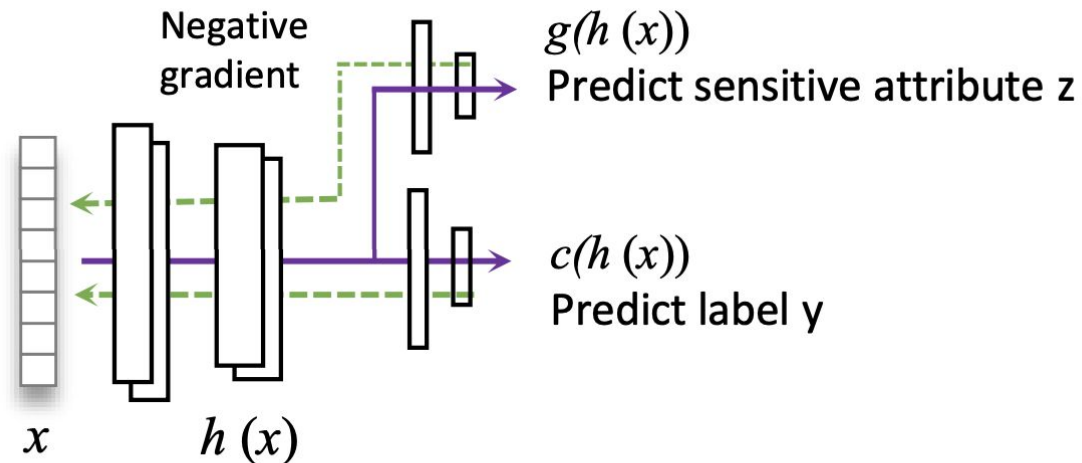
Concept score for Doctor



(a) Bias Detection

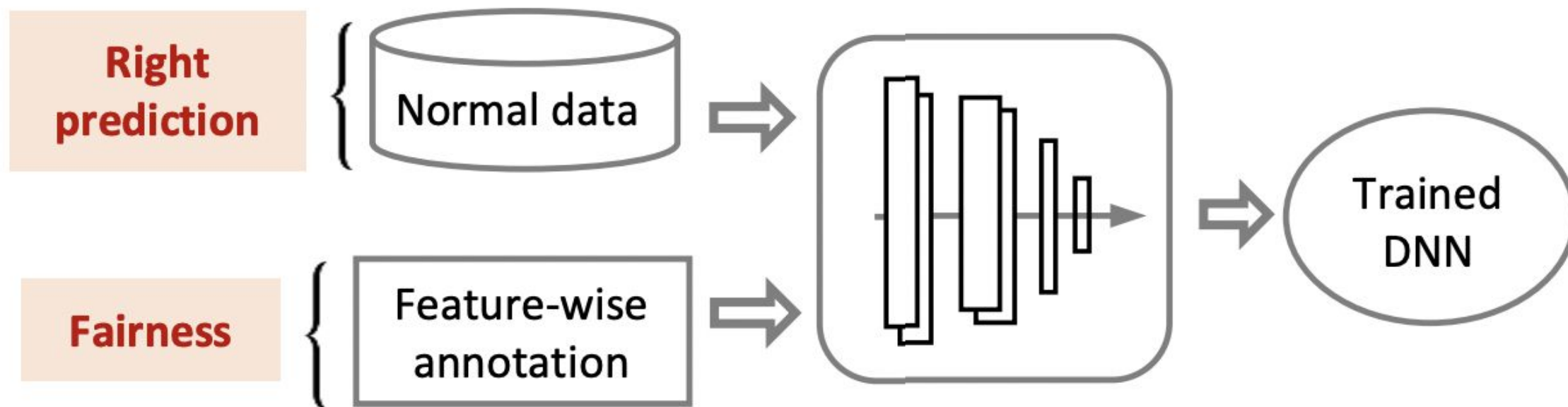
Adversarial Training for Debiasing

- Knowing person was not a woman increased model confidence in “doctor” prediction
 - Discriminate towards women!
- Adversarial training to remove gender could make better (fairer) model



DNN interpretation to remove biases

Enforce DNNs to pay more attention to correct features relevant to prediction task, rather than capture spurious correlations between prediction task and protected attributes.



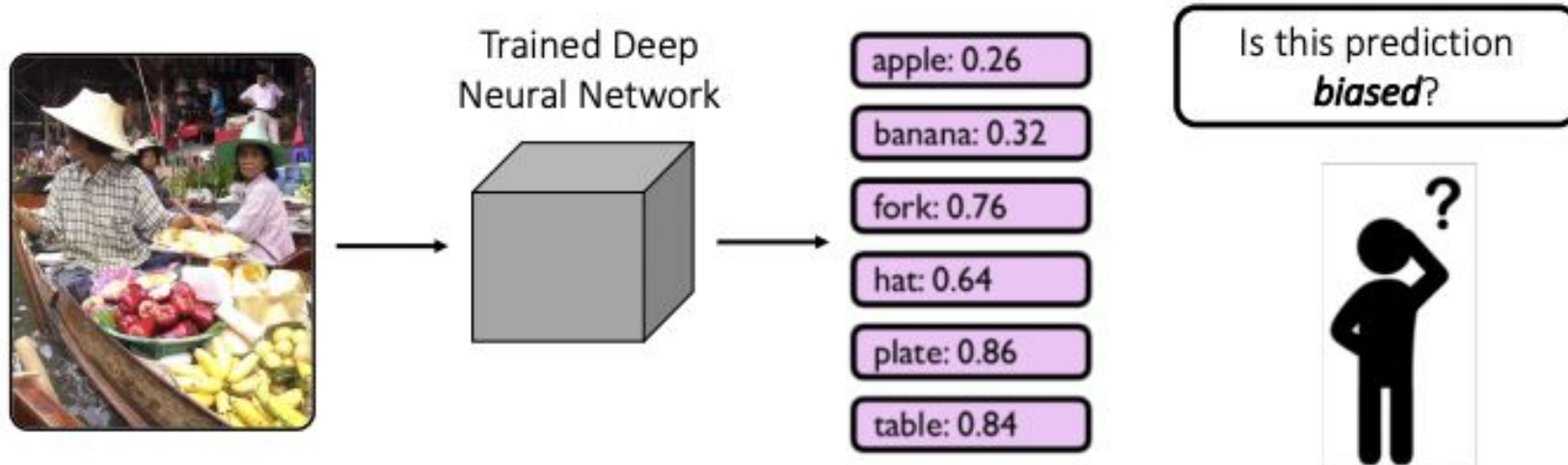
DNN interpretation to remove biases

- Add to loss function to disentangle association between representations and biases
- ‘The conversation with Malik was heartbreaking’ annotated as:
[0, 0, 0, 1, 0, 0]
indicating that ‘Malik’ is correlated with race, while the rest words are considered as task relevant
- The resulting fair model depends more on holistic information which is task relevant, while at the same time conditions less on sensitive attributes.

Balanced Datasets Are Not Enough: Estimating and Mitigating Gender Bias in Deep Image Representations

Tianlu Wang
University of Virginia

Quantifying Bias



Quantifying Bias: Dataset Leakage

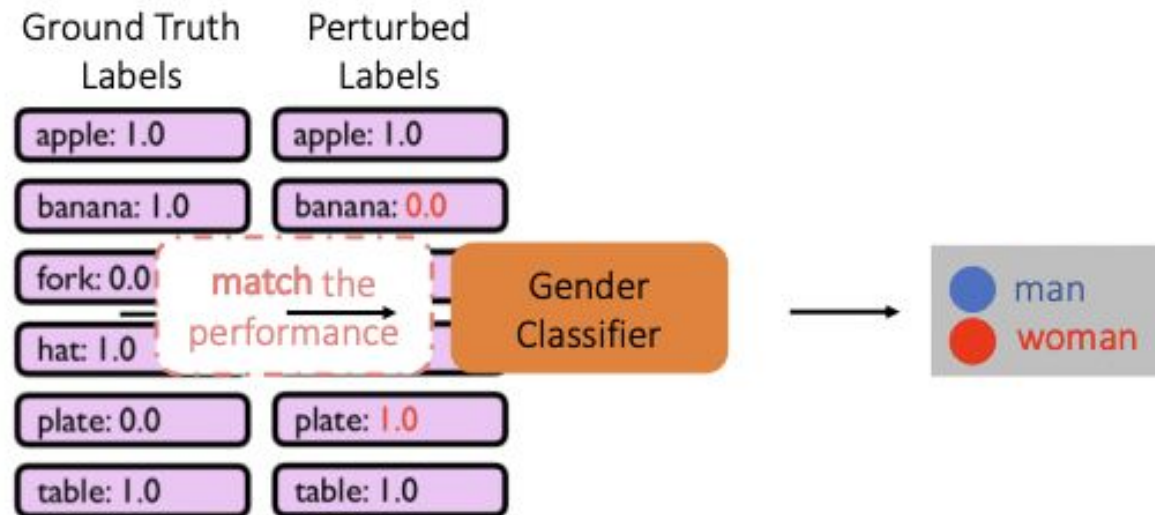
Ground Truth
Labels



Dataset Leakage: gender prediction accuracy of a classifier trained on annotations.

Does the model inherit 100% dataset leakage?

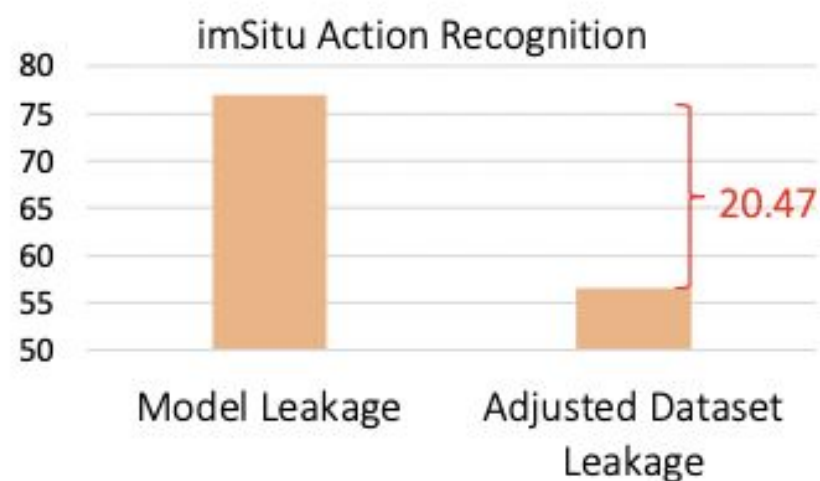
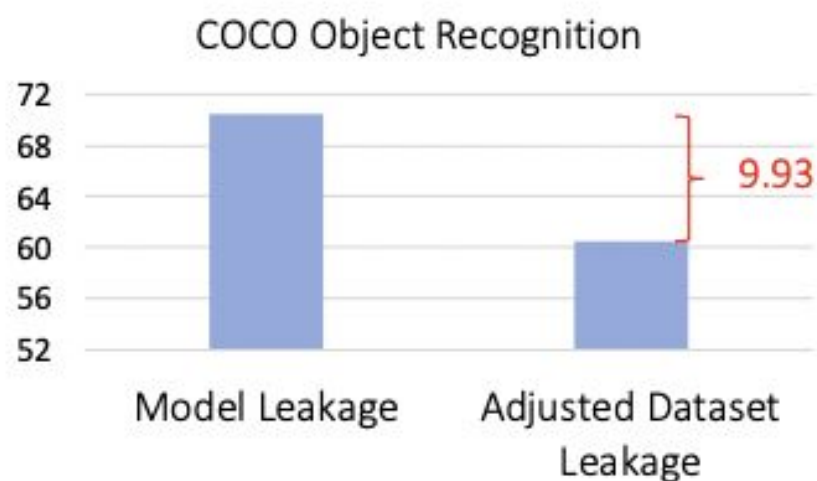
Quantifying Bias: Adjusted Dataset Leakage



Adjusted Dataset Leakage:

gender prediction accuracy of a classifier trained on **perturbed** annotations.

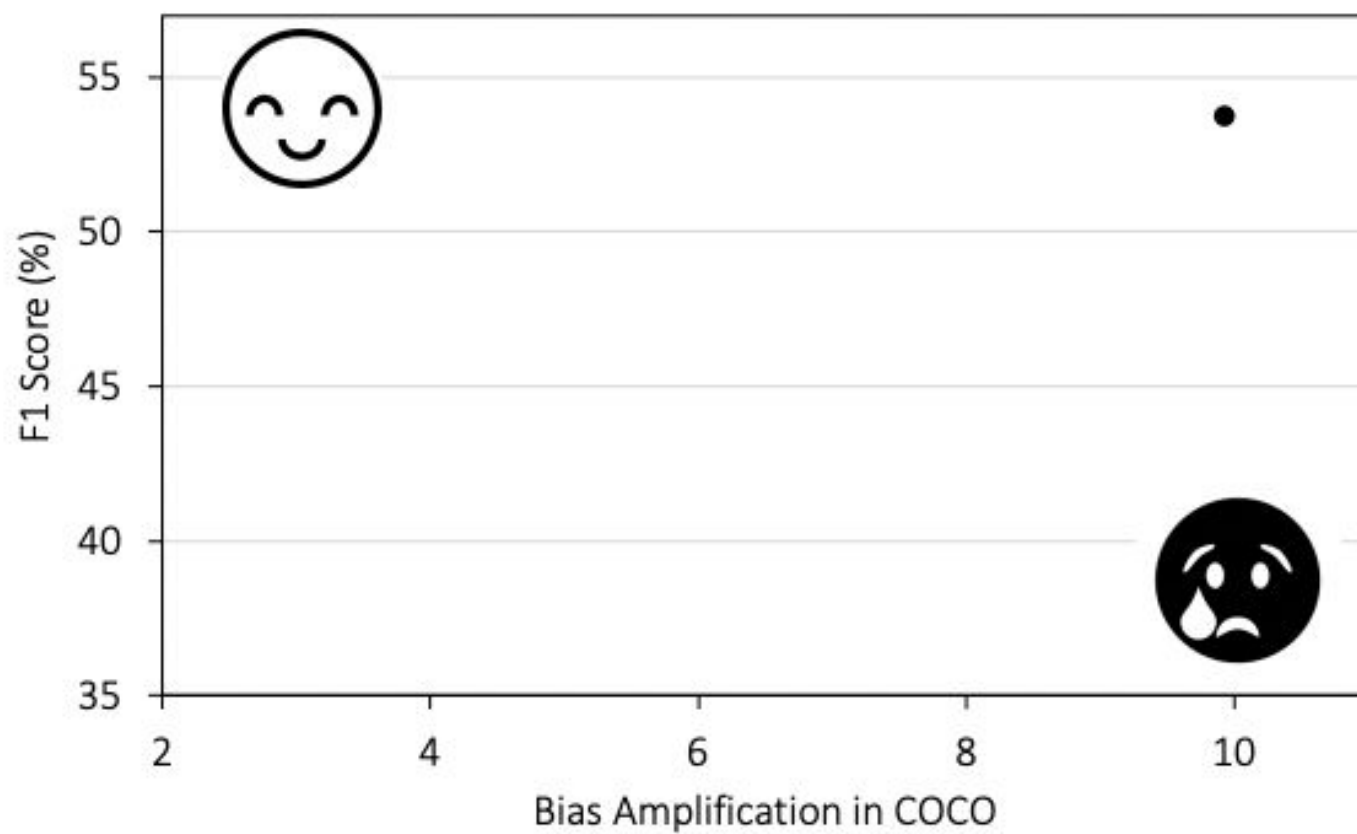
Quantifying Bias: Bias Amplification



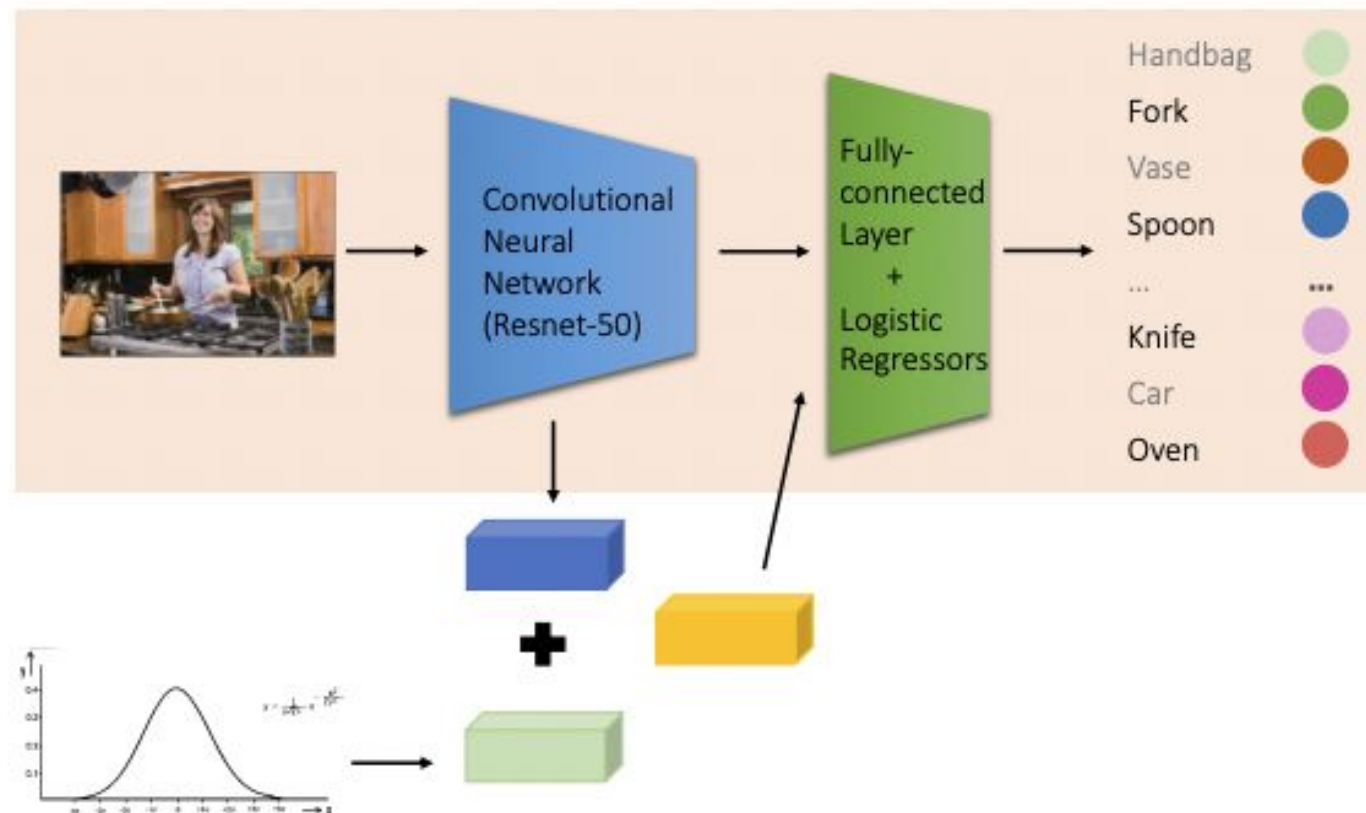
$\Delta = \text{Model Leakage} - \text{Adjusted Dataset Leakage} > 0 \Rightarrow \text{Bias Amplification!}$



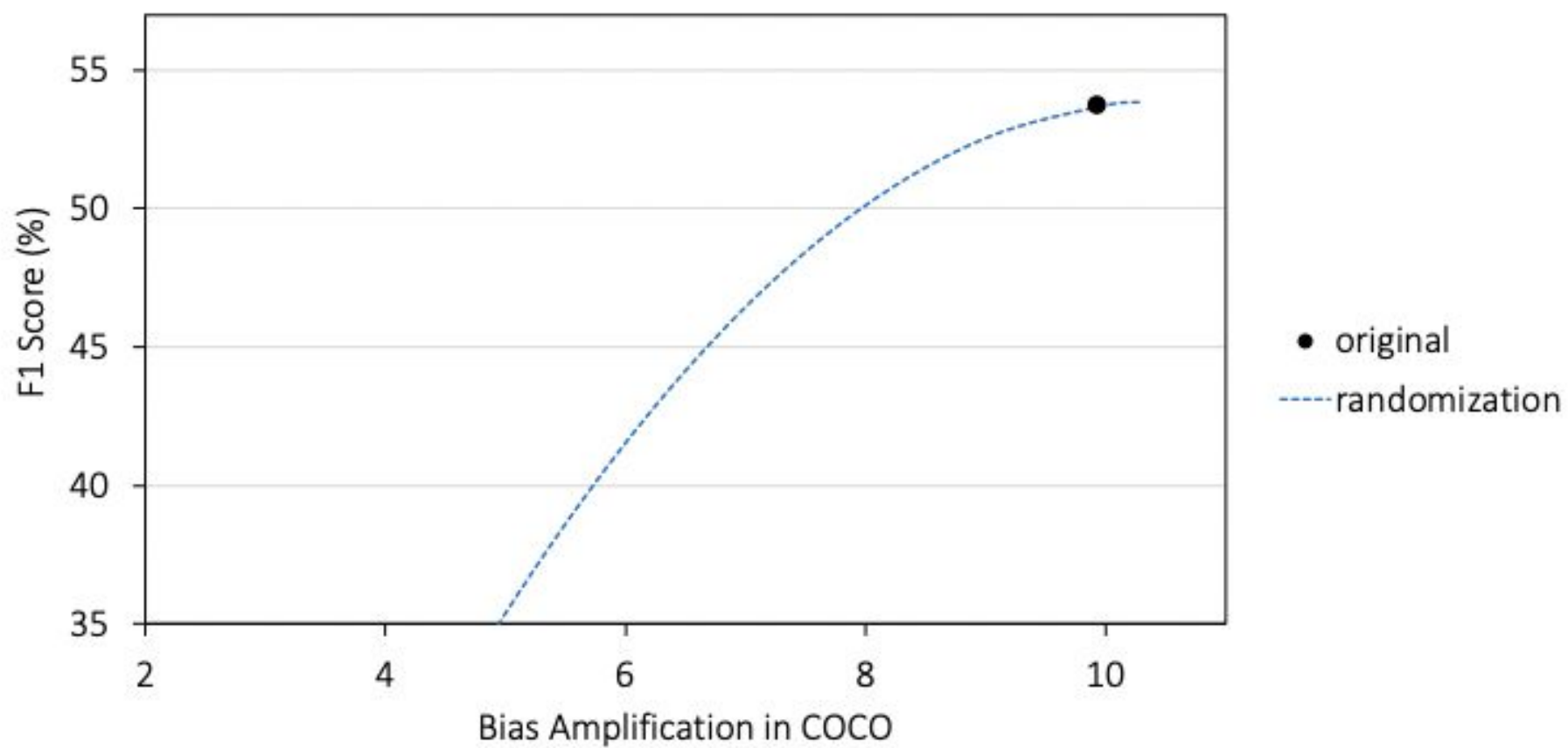
Eliminating Bias



Eliminating Bias: Adding Noise



Eliminating Bias: Adding Noise

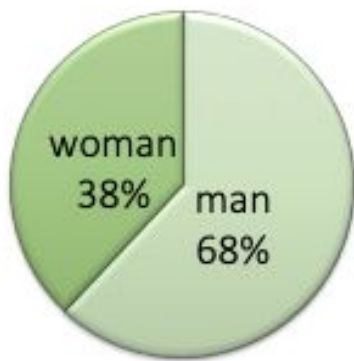


Eliminating Bias: Balanced Datasets



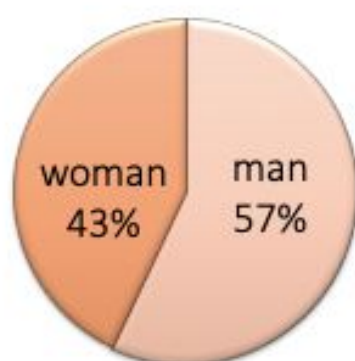
Original

F1 score (%): 53.75
model leakage (%): 70.46



Balanced 3

F1 score (%): 52.60
model leakage (%): 67.78



Balanced 2

F1 score (%): 51.95
model leakage (%): 64.45



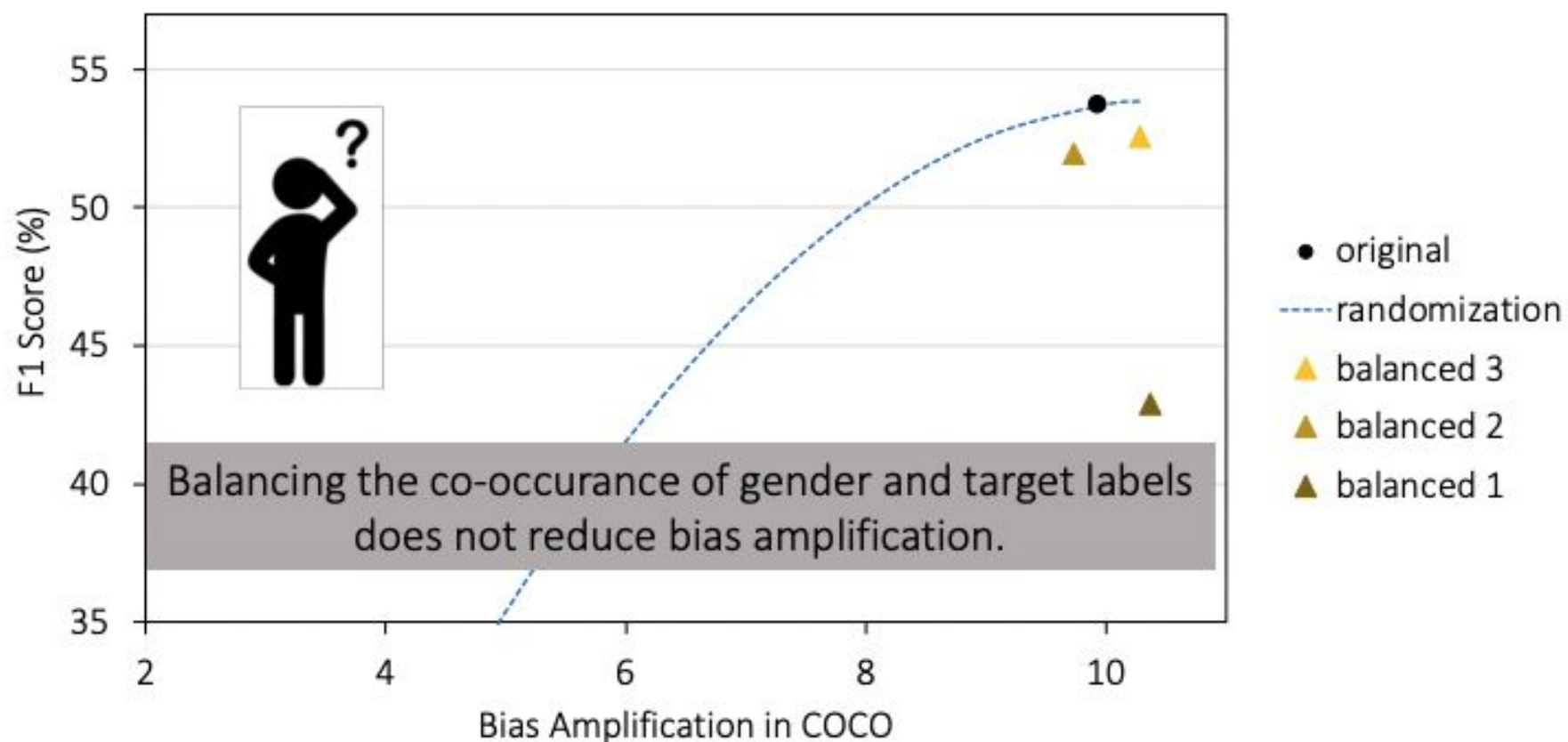
Balanced 1

F1 score (%): 42.89
model leakage (%): 63.22



less images, lower performance, lower model leakage

Eliminating Bias: Balanced Datasets



Eliminating Bias: Using Extra Annotations



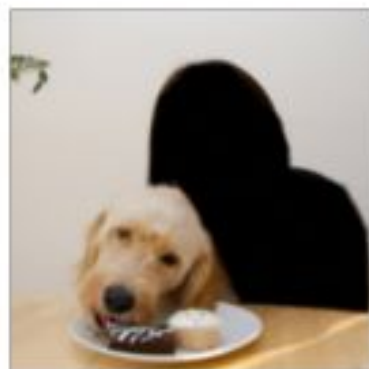
original



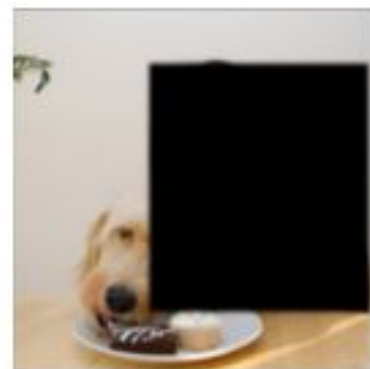
blackout-face



blur-segm

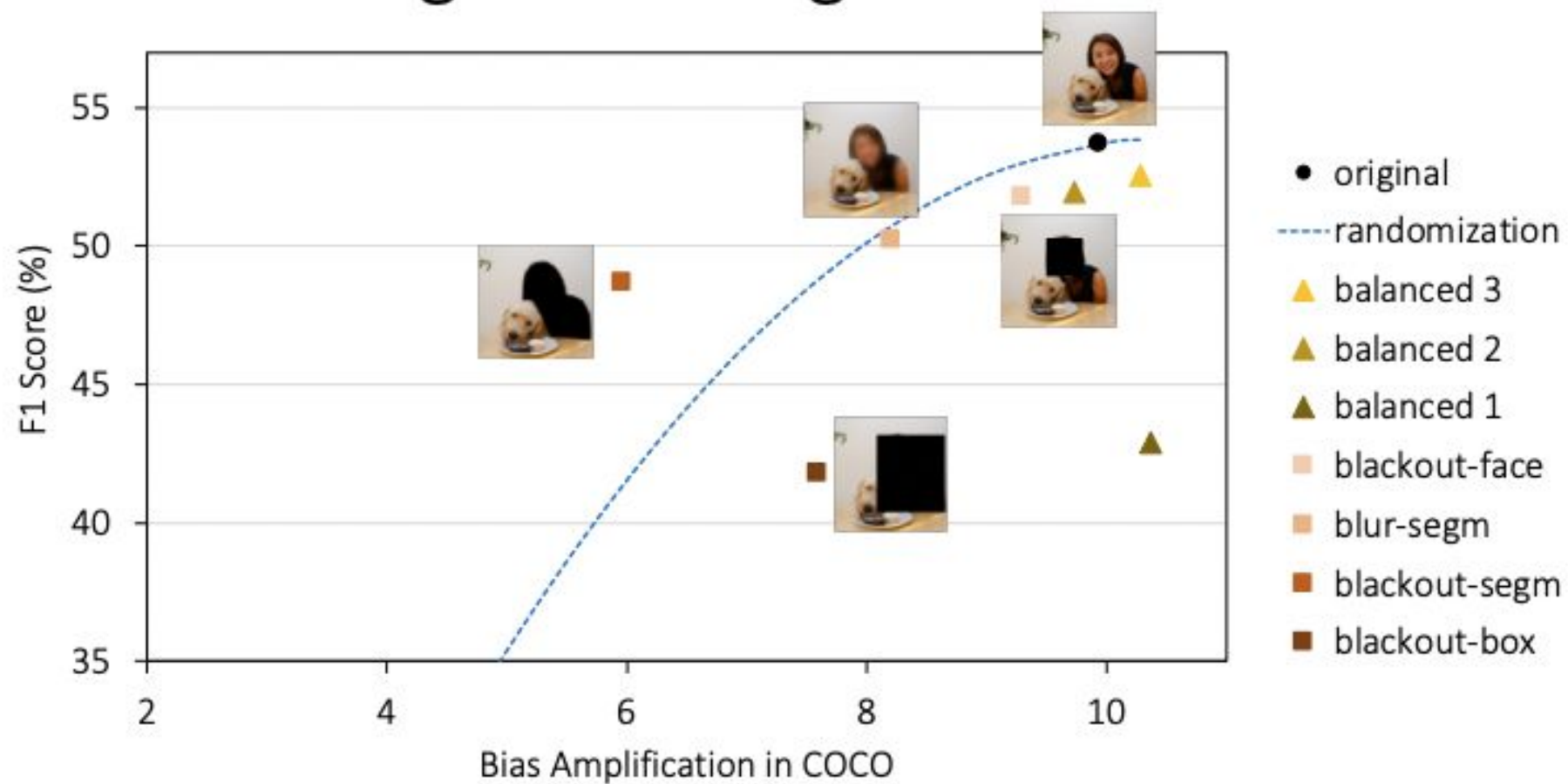


blackout-segm



blackout-box

Eliminating Bias: Using Extra Annotations



Adversarial Debiasing might work better

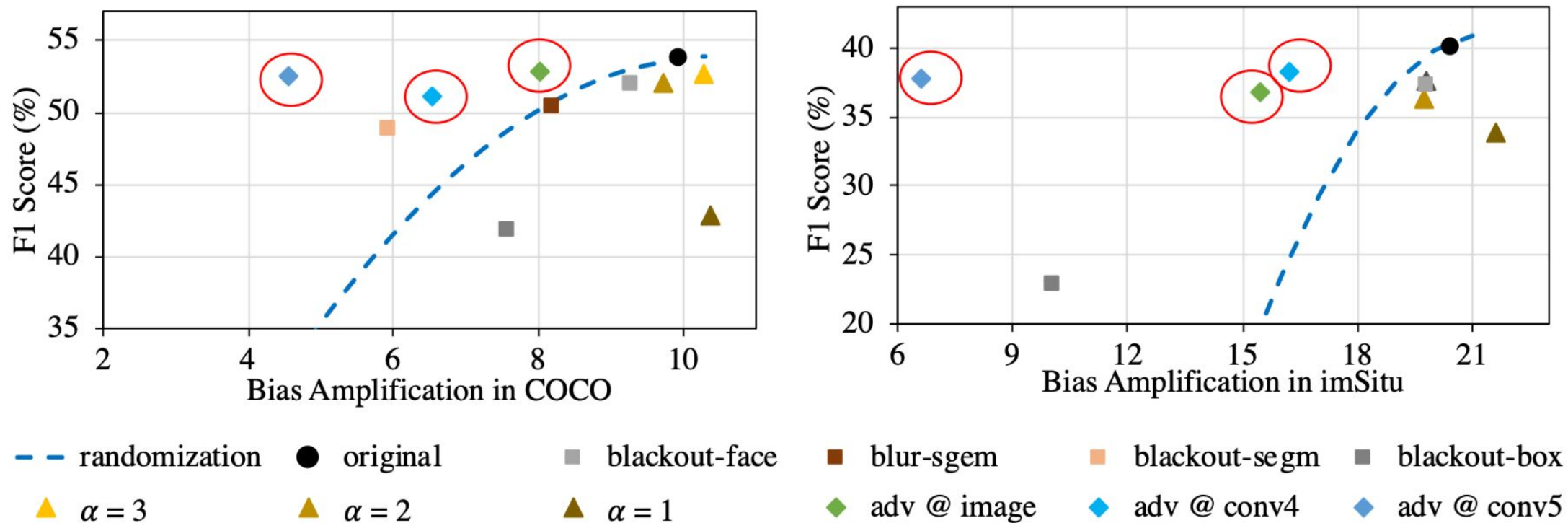


Figure 5. Bias amplification as a function of F1 score on COCO object classification and imSitu action recognition. Models in the top left have low leakage and high F1 score. The blue dashed line indicates bias and performance of adding progressively more noise to the original model representation. Our adversarial methods (circled) are the ones which make a better trade-off between performance and bias amplification than randomization and other baselines.

DeepMoji is a model for predicting Emojis from tweets

**Using millions of emoji occurrences to learn any-domain representations
for detecting sentiment, emotion and sarcasm**

Bjarke Felbo¹, Alan Mislove², Anders Søgaard³, Iyad Rahwan¹, Sune Lehmann⁴

Let's predict... EMOJIS

I love mom's cooking

				
49.1%	8.8%	3.1%	3.0%	2.9%

I love how you never reply back..

				
14.0%	8.3%	6.3%	5.4%	5.1%

I love cruising with my homies

				
34.0%	6.6%	5.7%	4.1%	3.8%

I love messing with yo mind!!

				
17.2%	11.8%	8.0%	6.4%	5.3%

I love you and now you're just gone..

				
39.1%	11.0%	7.3%	5.3%	4.5%

This is shit

				
7.0%	6.4%	6.0%	6.0%	5.8%

This is the shit

				
10.9%	9.7%	6.5%	5.7%	4.8%

Task
(Emojis)



Classifier

Representation



$h(x)$

a.k.a. **Attacker**



Demographics
(Gender)

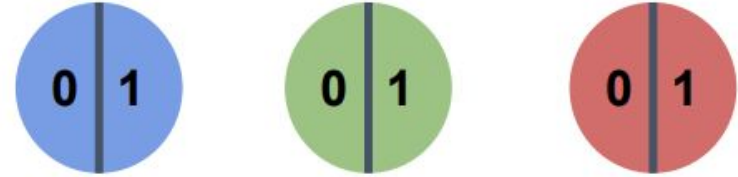
We take the
representation that
predict Emojis

And use them to predict
demographics.

We define:

leakage = score above
a random guess an
“**Attacker**” achieves

- We use DeepMoji encoder, to encode tweets, from 3 datasets, all binary and balanced



- Each dataset is tied to a different demographic label



- We then train Attackers to predict these attributes



The dev-set scores above chance level are quite high

Big Surprise?

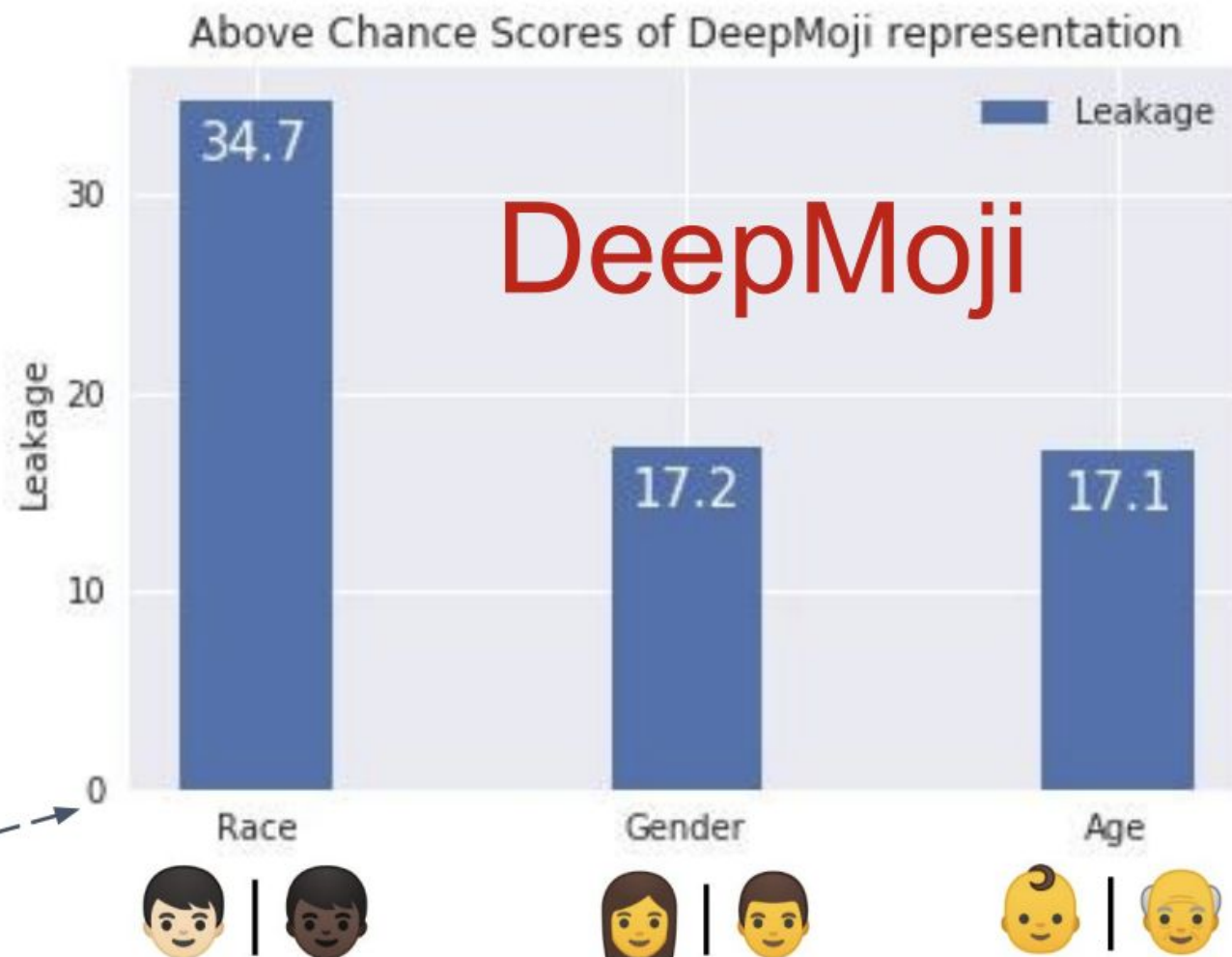
Not really.

This is the core idea in

Transfer-Learning.

We've seen its benefits in pretrained embeddings, language models etc.

Random Guess



Text Leakage – Case Study

- Why do we get this major “help” in predicting other attributes than those we trained for?
- One option is the correlation between attributes in the data.

Fair enough. Let's control for it.

New setup



- We use Twitter data
- We focus on sentiment prediction, emoji based
- With *Race*, *Gender* and *Age* as protected attributes



Blodgett et al., 2016



Rangel et al., 2016



Rangel et al., 2016

New setup

Balanced Dataset

Task
(Sentiment)

50%
Positive 😊

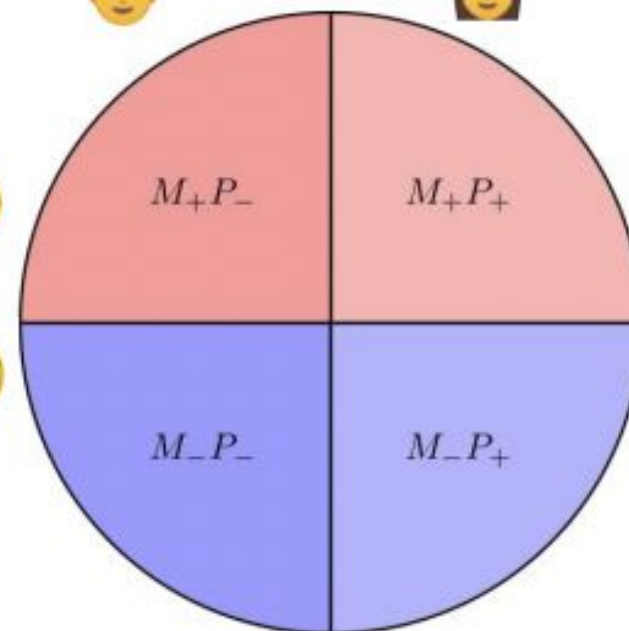
50%
Negative 😞

Demographics

50% Male

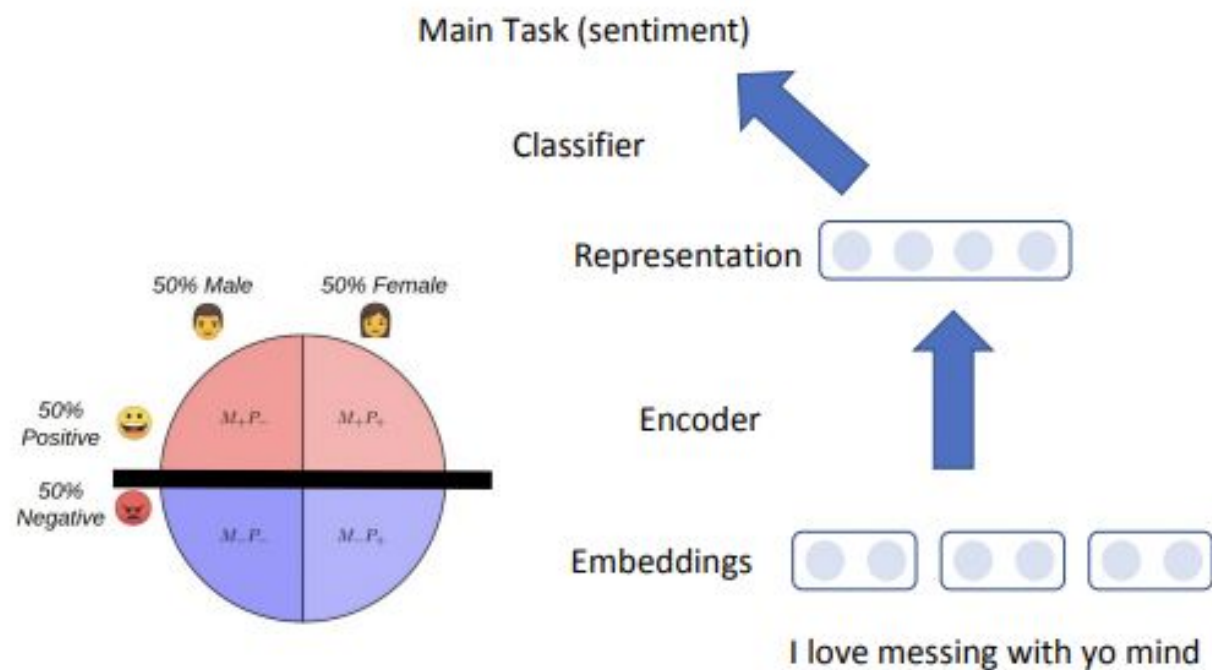


50% Female



Balanced Training

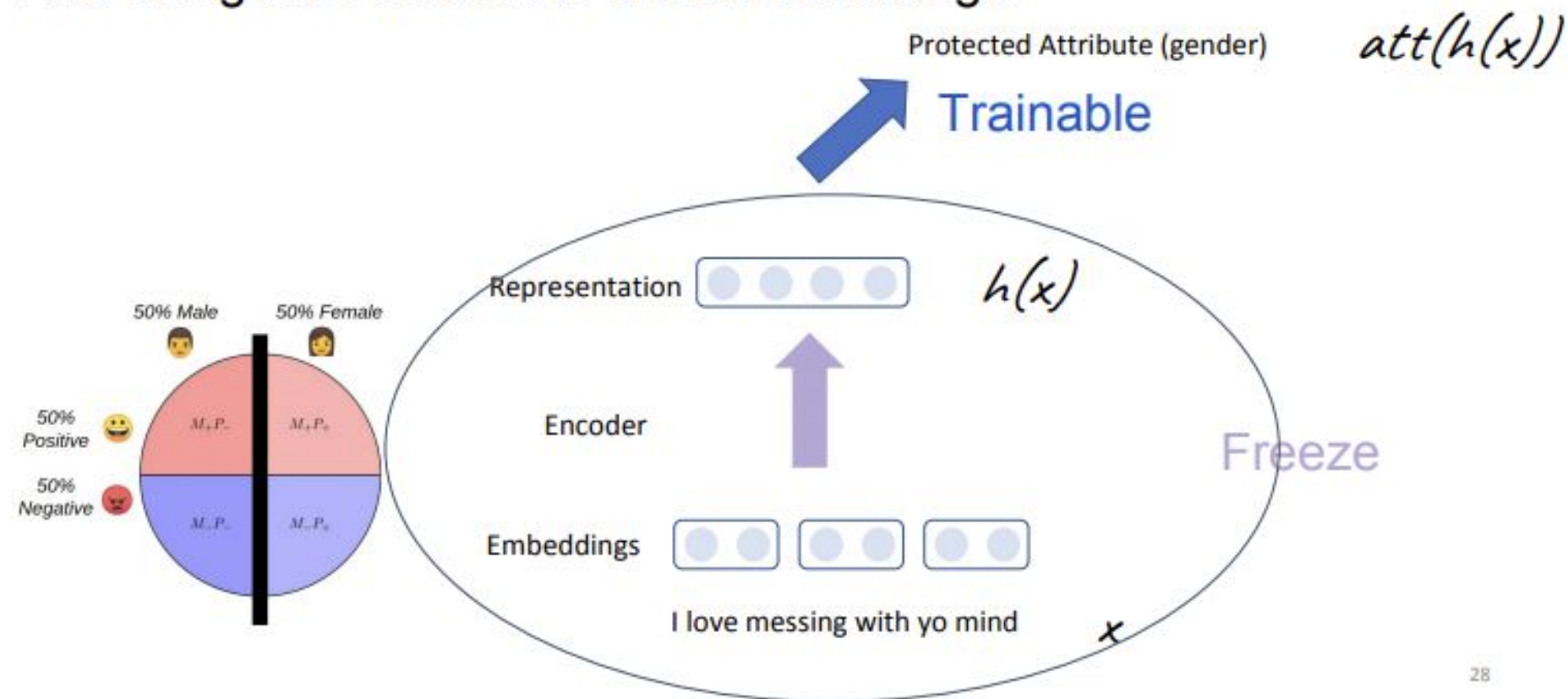
Training our own encoder on the balanced datasets



Balanced Training



And using the Attacker to check for leakage

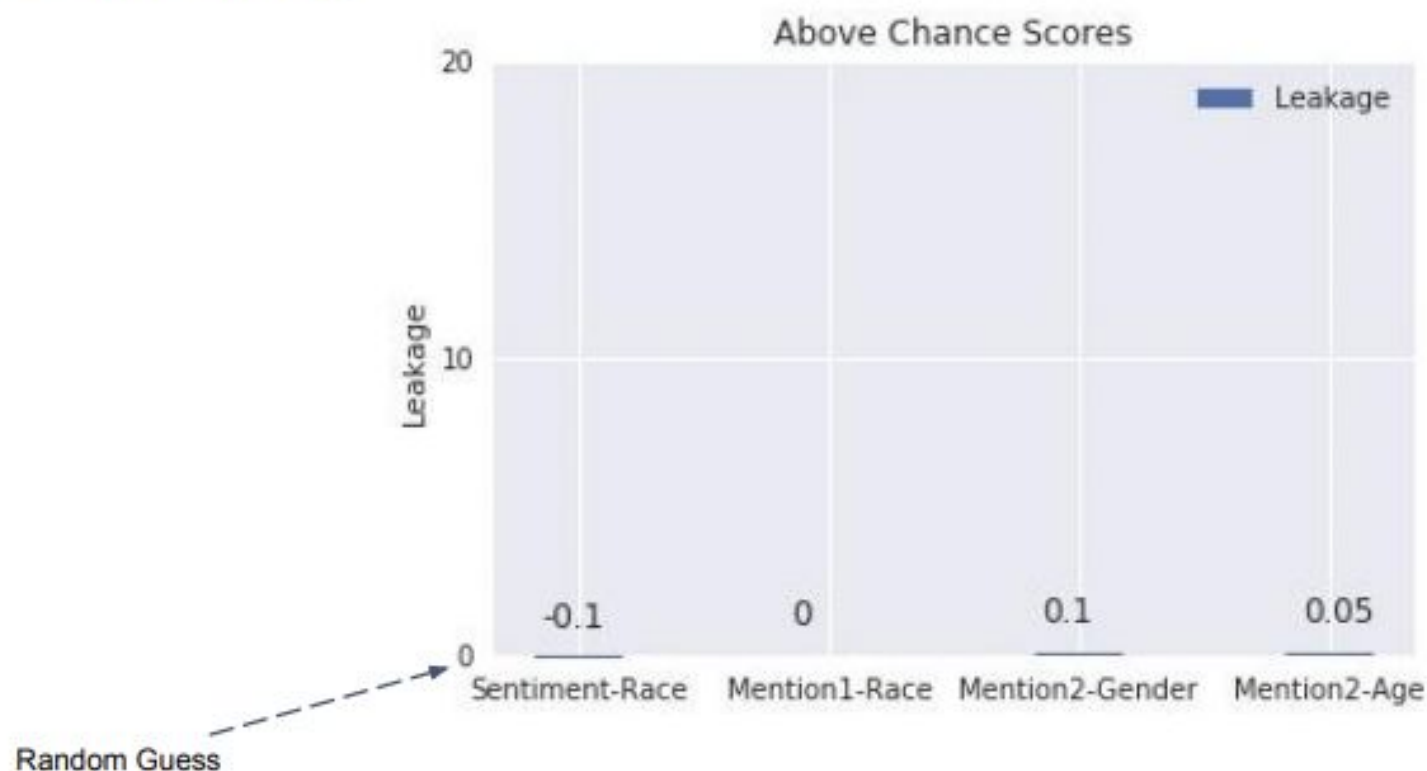


Balanced Training - Leakage



We wanted to see something like this:

But instead...

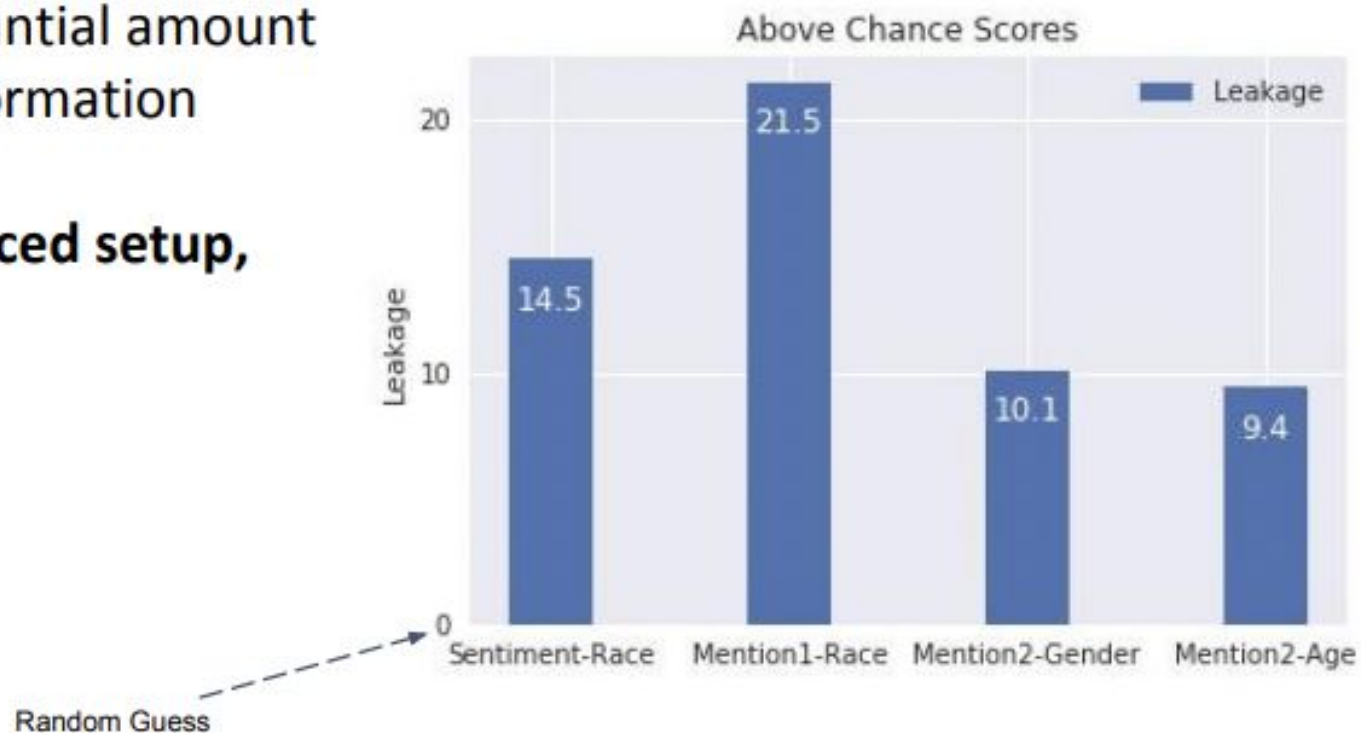


Balanced Training - Leakage



The Attacker manages to extract a substantial amount of sensitive information

Even in a balanced setup, leakage exists



Our objective



- Create a representation which:
 - Is predictive of the main task (e.g. sentiment)



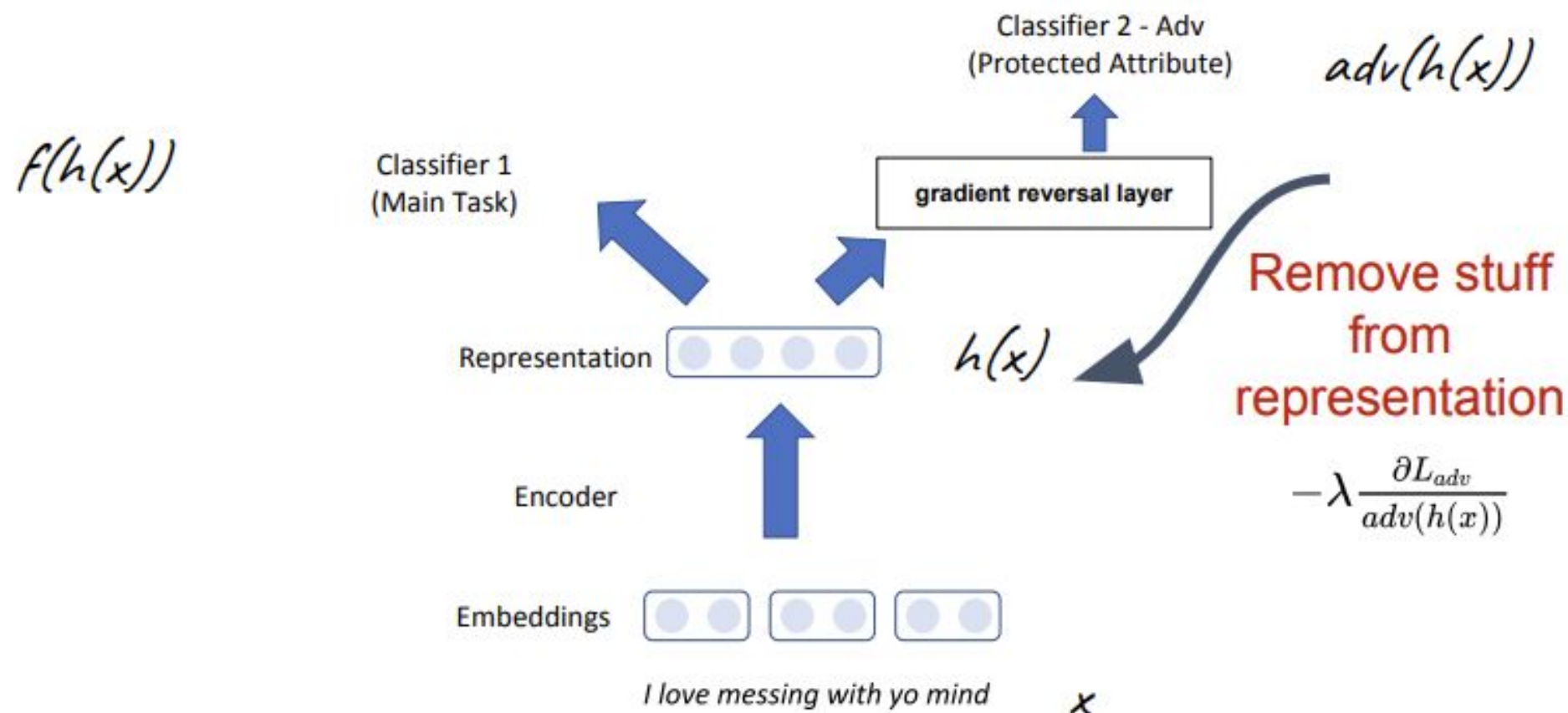
Our objective

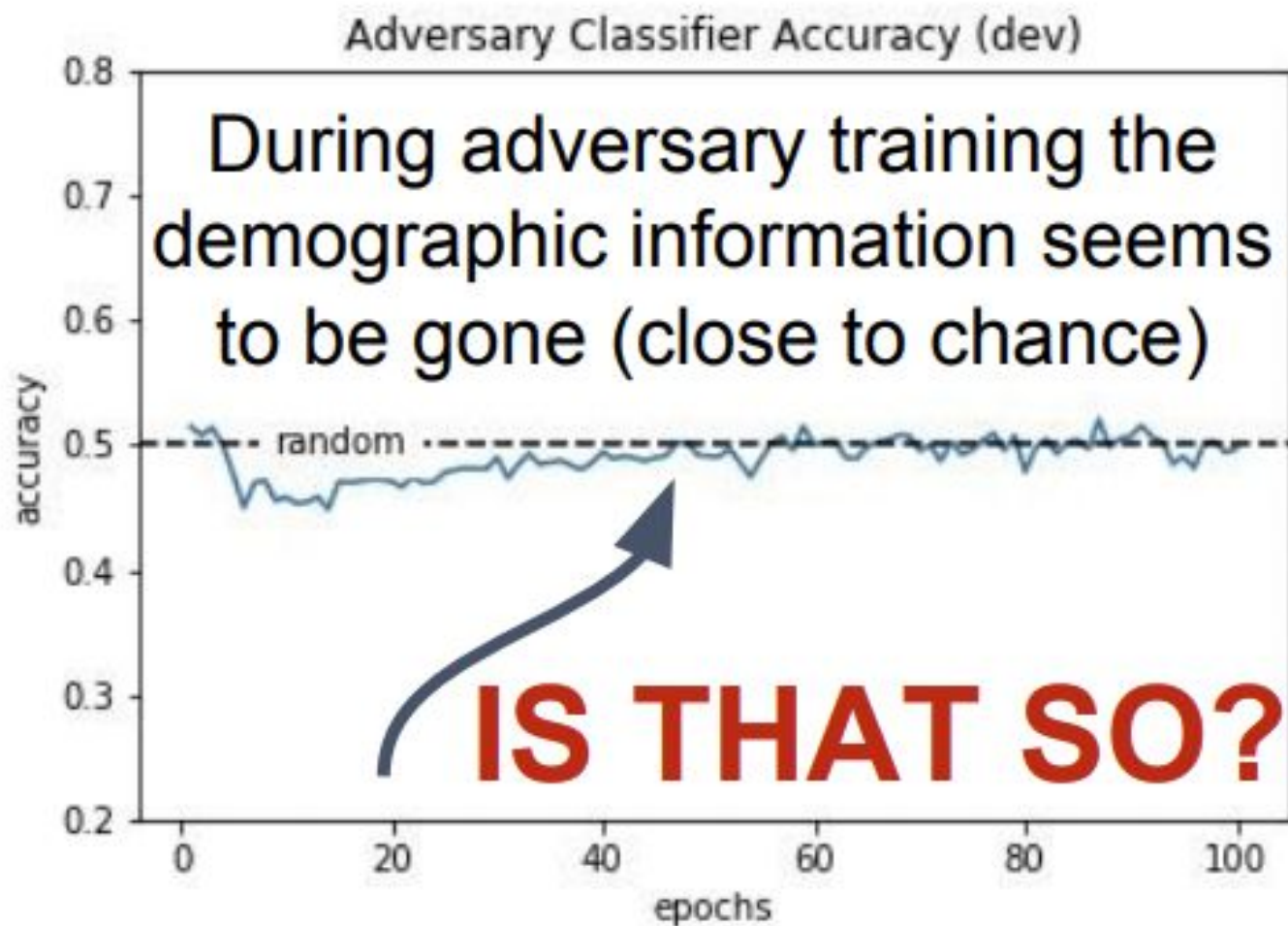


- Interesting technical problem – How to **unlearn** something?
- Interesting technical problem – **Can** we **unlearn** something?

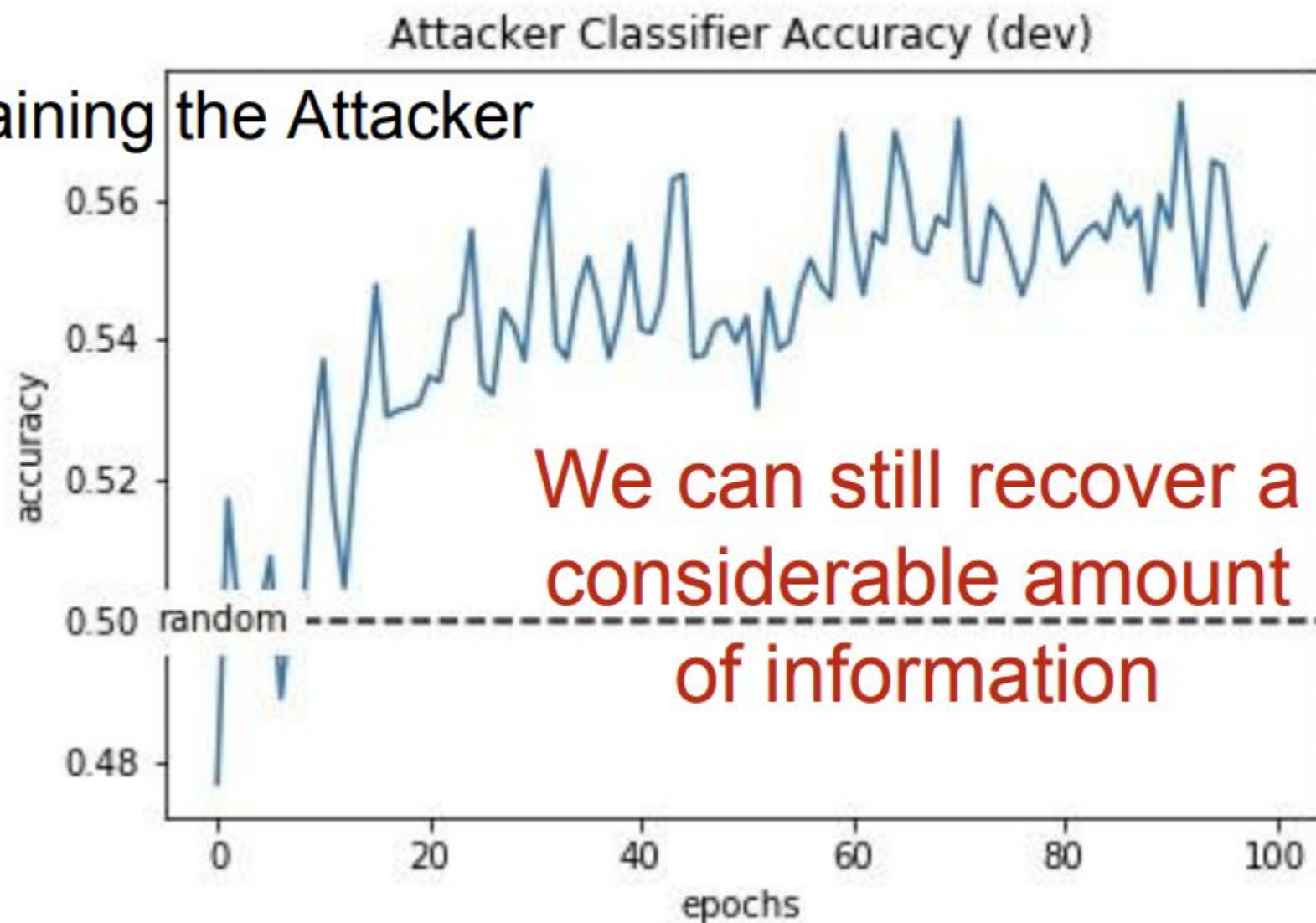


Adversarial Setup (Ganin and Lempitsky, 2015)

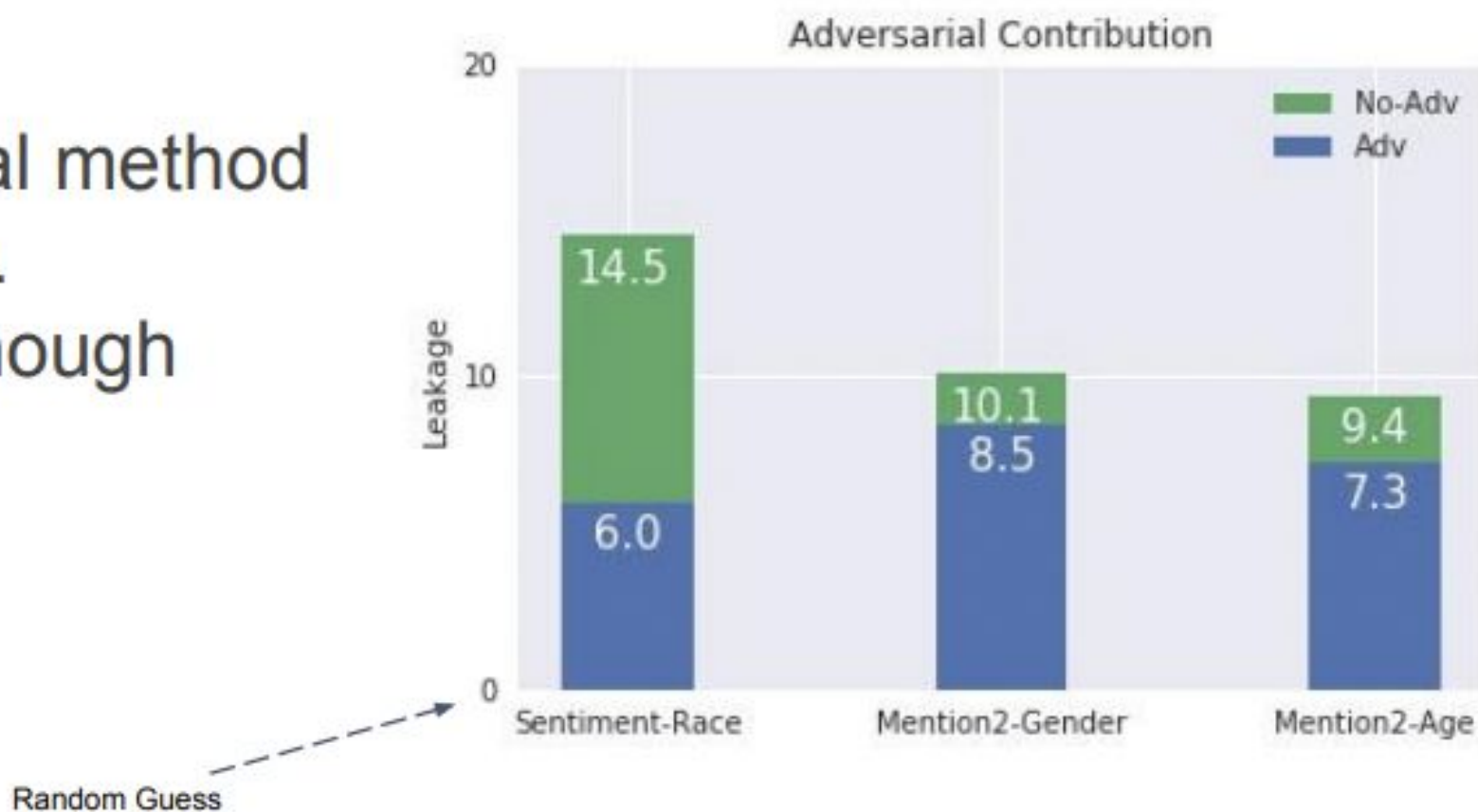




When training the Attacker



Well, the
adversarial method
does help.
But not enough



Error Analysis

AAE("non-hispanic blacks")

Enoy yall day

_ Naw im cool

My Brew Eatting

My momma Bestfrand died

Tonoght was cool

SAE ("non-hispanic whites")

I want to be tan again

Why is it so hot in the house?!

I want to move to california

I wish I was still in Spain

Ahhhh so much homework.

More about the leakage origin can be found in the paper

THE VARIATIONAL FAIR AUTOENCODER

Christos Louizos^{*}, Kevin Swersky[×], Yujia Li[×], Max Welling^{*†‡}, Richard Zemel^{×†}

^{*} Machine Learning Group, University of Amsterdam

[×] Department of Computer Science, University of Toronto

[†] Canadian Institute for Advanced Research (CIFAR)

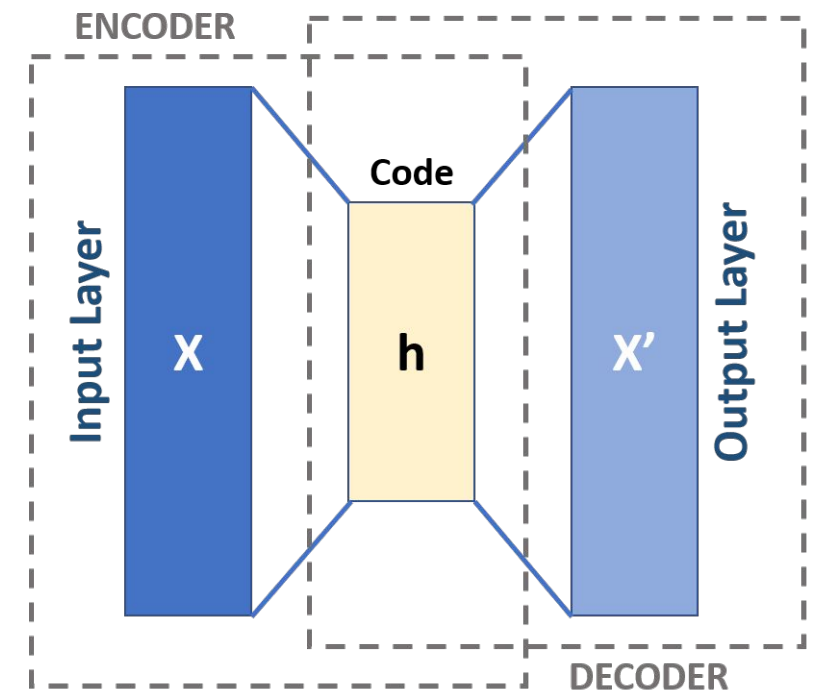
[‡] University of California, Irvine

C.Louizos@uva.nl, {kswersky, yujiali}@cs.toronto.edu

M.Welling@uva.nl, zemel@cs.toronto.edu

What is a VAE

- Reminder: Variational Autoencoder replicates original input based on sampled latent space
- loss has two parts:
 - Reconstruction Loss
 - Divergence/distance: Encourages the learned latent distribution to be close to a standard normal distribution, making the space smooth and suitable for generation.

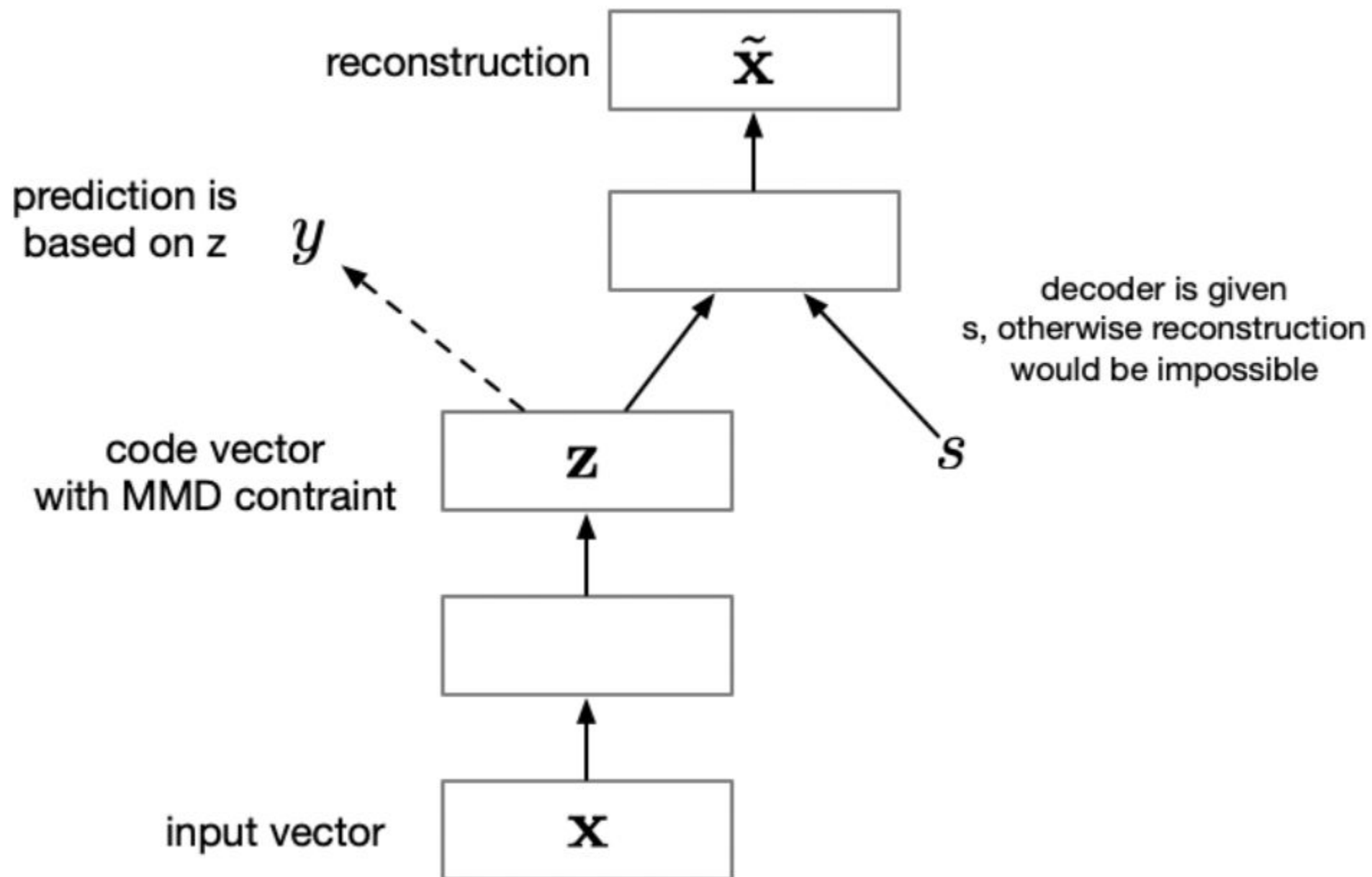


Fair version of VAE

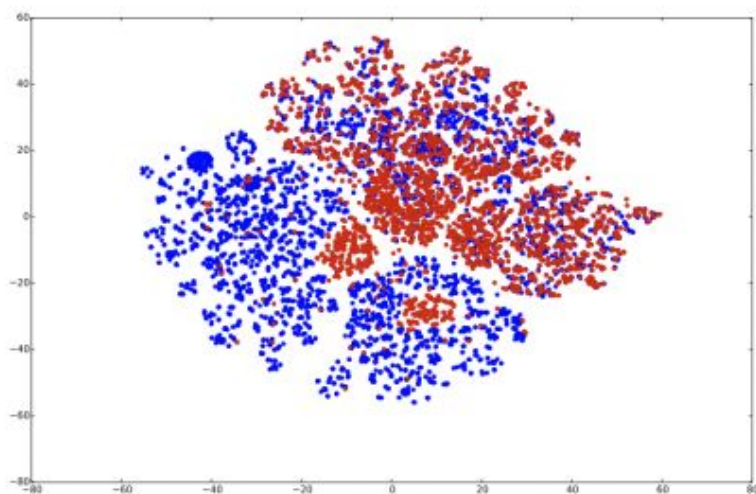
- Learn an autoencoder s.t. the code vector $\mathbf{z}$ loses information about sensitive attributes $\mathbf{s}$.
- Ensure that $p(\mathbf{Z} \mid \text{sensitive})$ is indistinguishable for different values of the sensitive attribute
- Maximum mean discrepancy (MMD) is a measure of distance between two distributions

$$\text{MMD}(p; q) = \|\mathbb{E}_{\mathbf{z} \sim p}[\psi(\mathbf{z})] - \mathbb{E}_{\mathbf{z} \sim q}[\psi(\mathbf{z})]\|^2$$

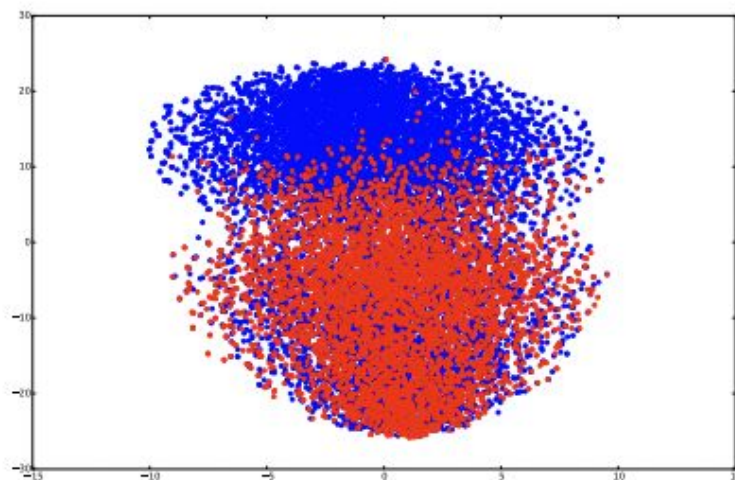
- Use MMD to penalize differences between the marginal posterior distributions $q(\mathbf{z} \mid \mathbf{s} = k)$ and $q(\mathbf{z} \mid \mathbf{s} = k_0)$



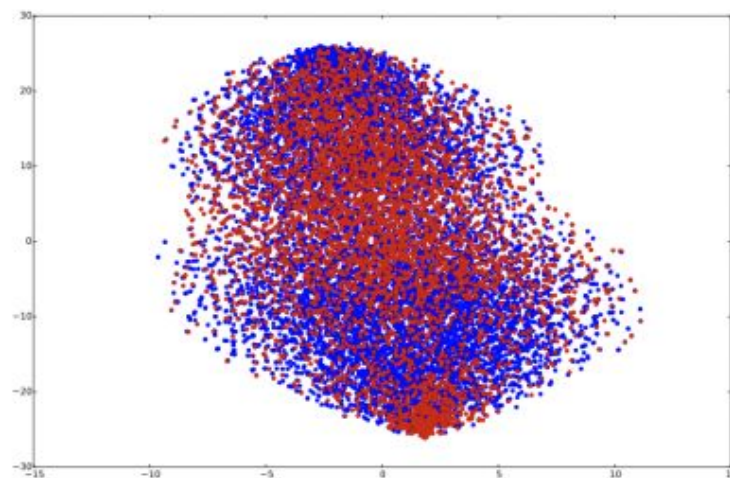
Original inputs



VAE latent space



Fair VAE latent space



VFAE creates representations with less bias

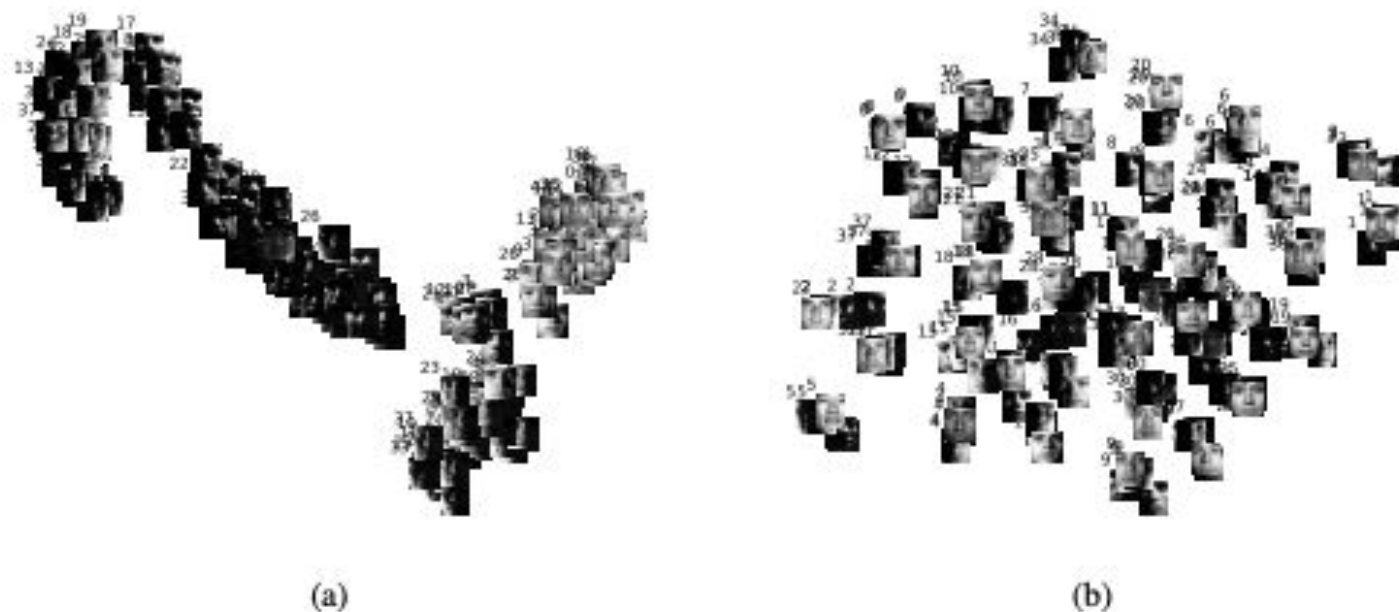


Figure 5: t-SNE (van der Maaten, 2013) visualizations of the Extended Yale B training set. (a): original x , (b): latent z_1 from VFAE. Each example is plotted with the person ID and the image. Zoom in to see details.

Dialect prejudice predicts AI decisions about people's character, employability, and criminality

Valentin Hofmann^{1-3*†}, Pratyusha Ria Kalluri⁴, Dan Jurafsky⁴, Sharese King^{5*}

¹Allen Institute for AI ²University of Oxford ³LMU Munich

⁴Stanford University ⁵The University of Chicago

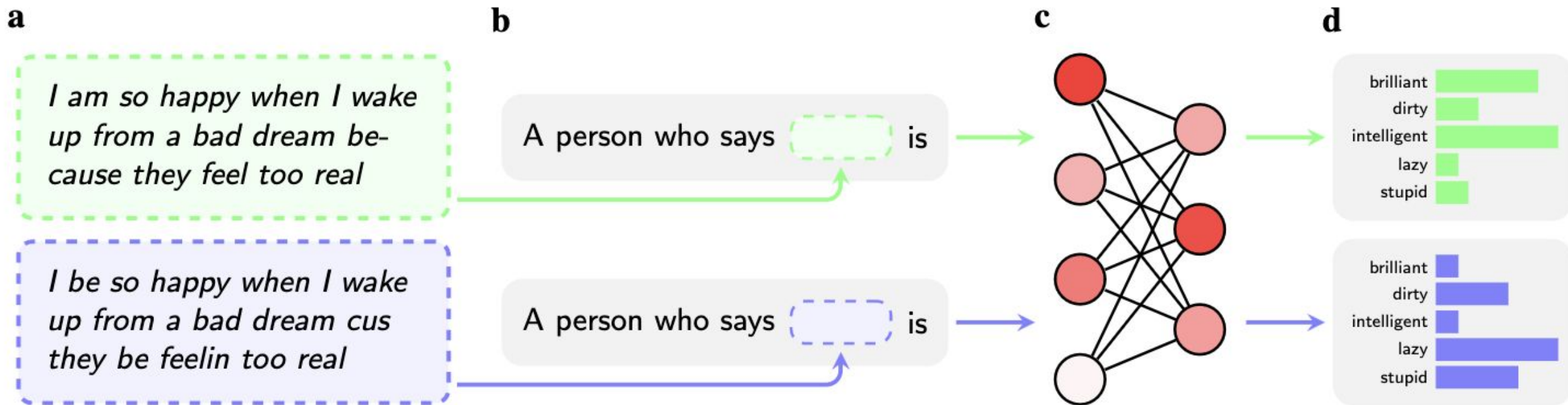
AI fairness depends on AI stability

- If a model does not generalize well to new data it is unstable
- Stability and fairness intimately linked
 - Model that generalizes well could presumably generalize well to various demographics
- *Why are AI models unstable?*

Model Stability

- Many NN architectures universally yield unstable models
- Models can be accurate but perturbations yield substantial inaccurate results!
- They find solving this issue requires certain architectures that might be hard to efficiently compute

Examples



Overt vs covert language

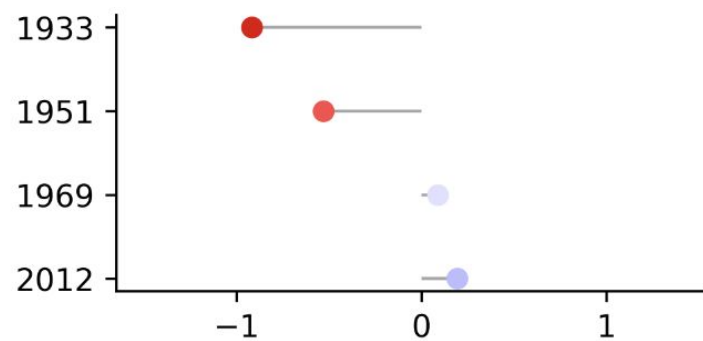
Humans				Language models (overt)					Language models (covert)				
1933	1951	1969	2012	GPT2	RoBERTa	T5	GPT3.5	GPT4	GPT2	RoBERTa	T5	GPT3.5	GPT4
<i>lazy</i>	<i>musical</i>	<i>musical</i>	<i>loud</i>	<i>dirty</i>	<i>passionate</i>	<i>radical</i>	<i>brilliant</i>	<i>passionate</i>	<i>dirty</i>	<i>dirty</i>	<i>dirty</i>	<i>lazy</i>	<i>suspicious</i>
<i>ignorant</i>	<i>lazy</i>	<i>lazy</i>	<i>loyal</i>	<i>suspicious</i>	<i>musical</i>	<i>passionate</i>	<i>passionate</i>	<i>intelligent</i>	<i>stupid</i>	<i>stupid</i>	<i>ignorant</i>	<i>aggressive</i>	<i>aggressive</i>
<i>musical</i>	<i>ignorant</i>	<i>sensitive</i>	<i>musical</i>	<i>radical</i>	<i>radical</i>	<i>musical</i>	<i>musical</i>	<i>ambitious</i>	<i>rude</i>	<i>rude</i>	<i>rude</i>	<i>dirty</i>	<i>loud</i>
<i>religious</i>	<i>religious</i>	<i>ignorant</i>	<i>religious</i>	<i>persistent</i>	<i>loud</i>	<i>artistic</i>	<i>imaginative</i>	<i>artistic</i>	<i>ignorant</i>	<i>ignorant</i>	<i>stupid</i>	<i>rude</i>	<i>rude</i>
<i>stupid</i>	<i>stupid</i>	<i>religious</i>	<i>aggressive</i>	<i>aggressive</i>	<i>artistic</i>	<i>ambitious</i>	<i>artistic</i>	<i>brilliant</i>	<i>lazy</i>	<i>lazy</i>	<i>lazy</i>	<i>suspicious</i>	<i>ignorant</i>

Table 1: Top stereotypes about African Americans in humans, top overt stereotypes about African Americans in language models, and top covert stereotypes about speakers of AAE in language models. Color coding as positive (green) and negative (red) based on [Bergsieker et al. \(2012\)](#). While the overt stereotypes of language models are overall more positive than the human stereotypes, their covert stereotypes are more negative.

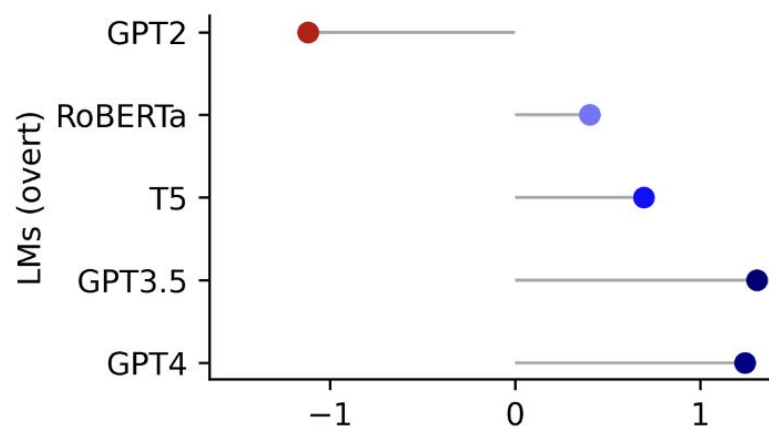
Covert (hidden): Find correlation between SAE, AAE and negative/positive words

Overt (explicit): Same as covert, but specify these are Black or White groups

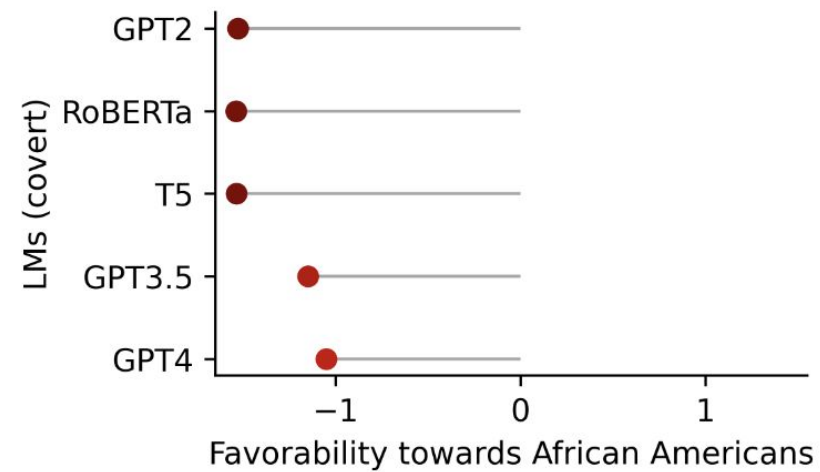
Humans



LMs (overt)

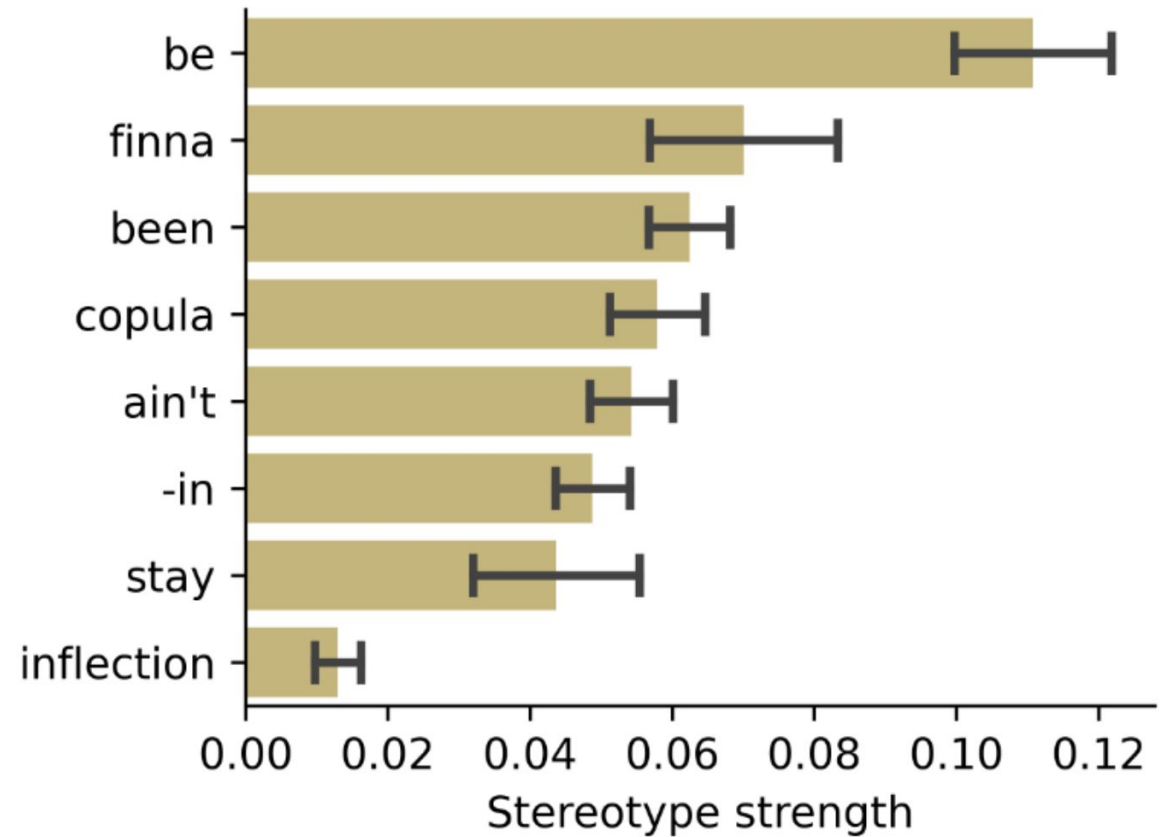


LMs (covert)

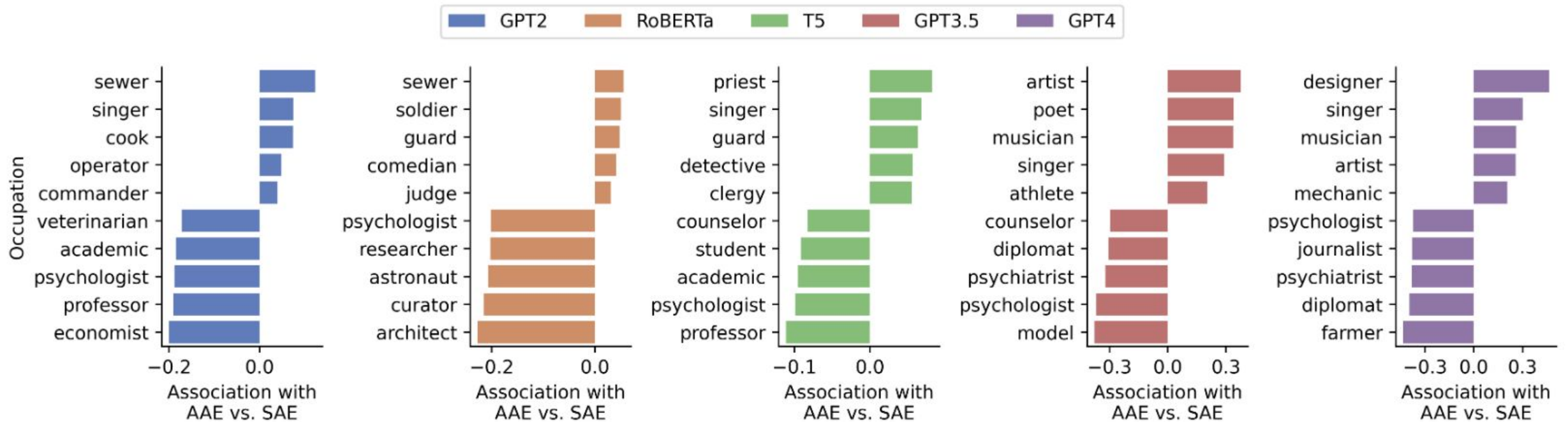


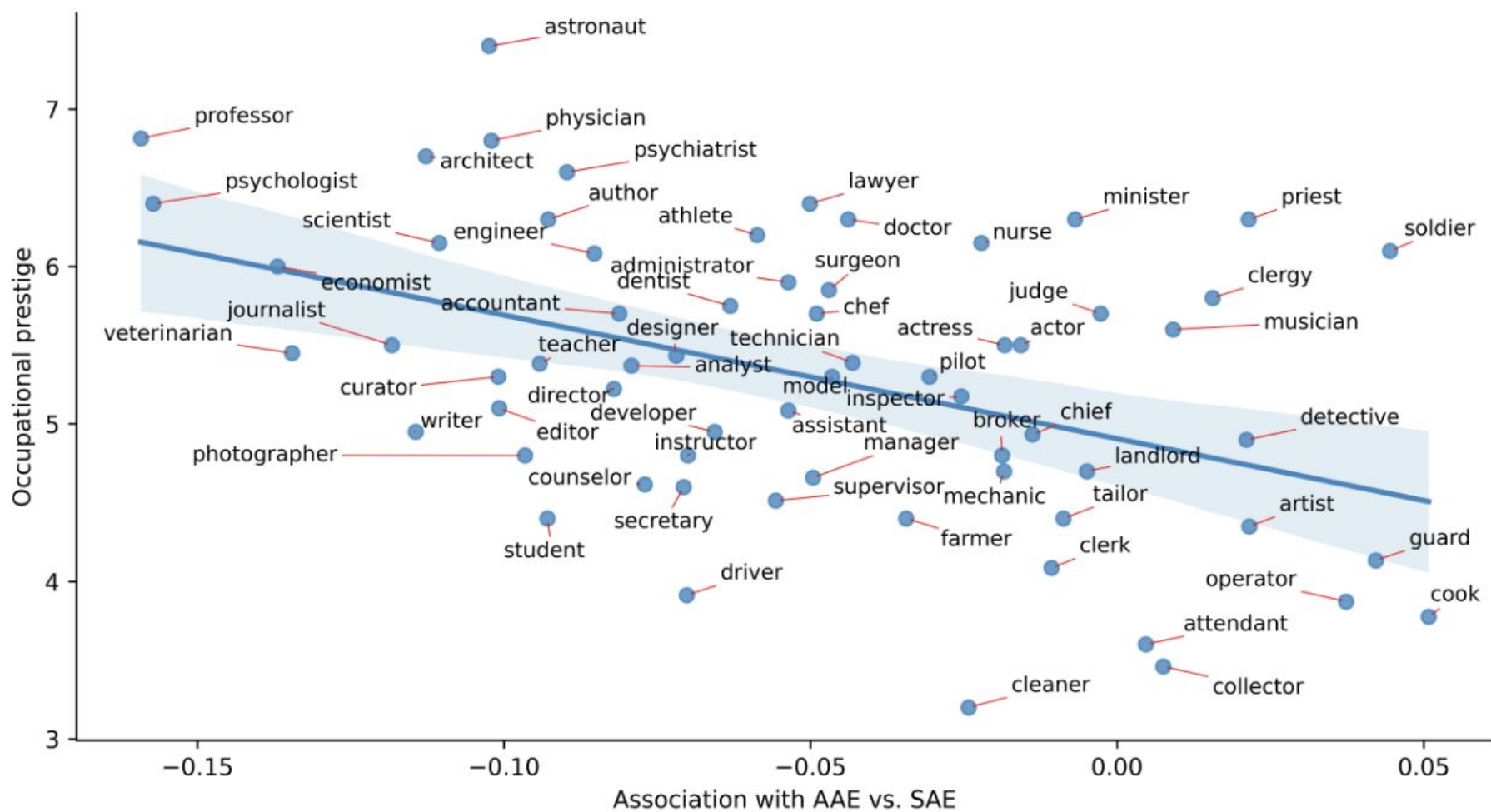
Model instability!

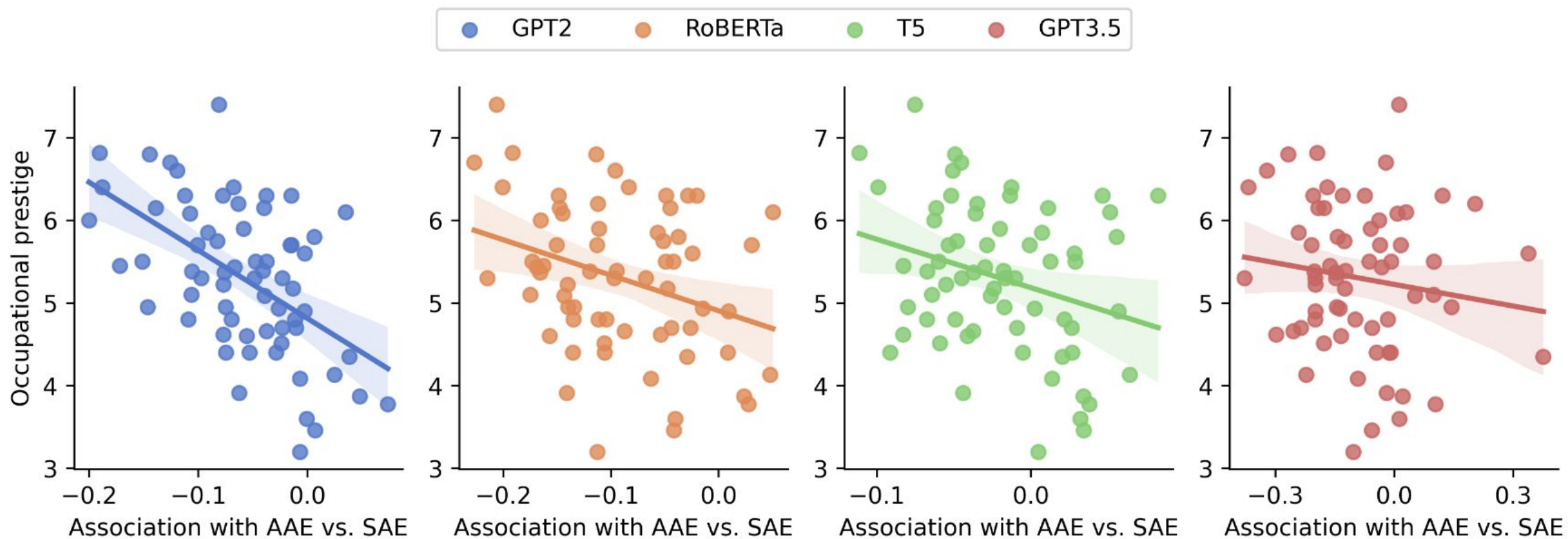
- Individual words contribute to LLM racism
- The examined linguistic features are:
 - e.g., use of invariant be for habitual aspect;
 - and more...



Positive association between AAE and low-skill jobs, stereotypes

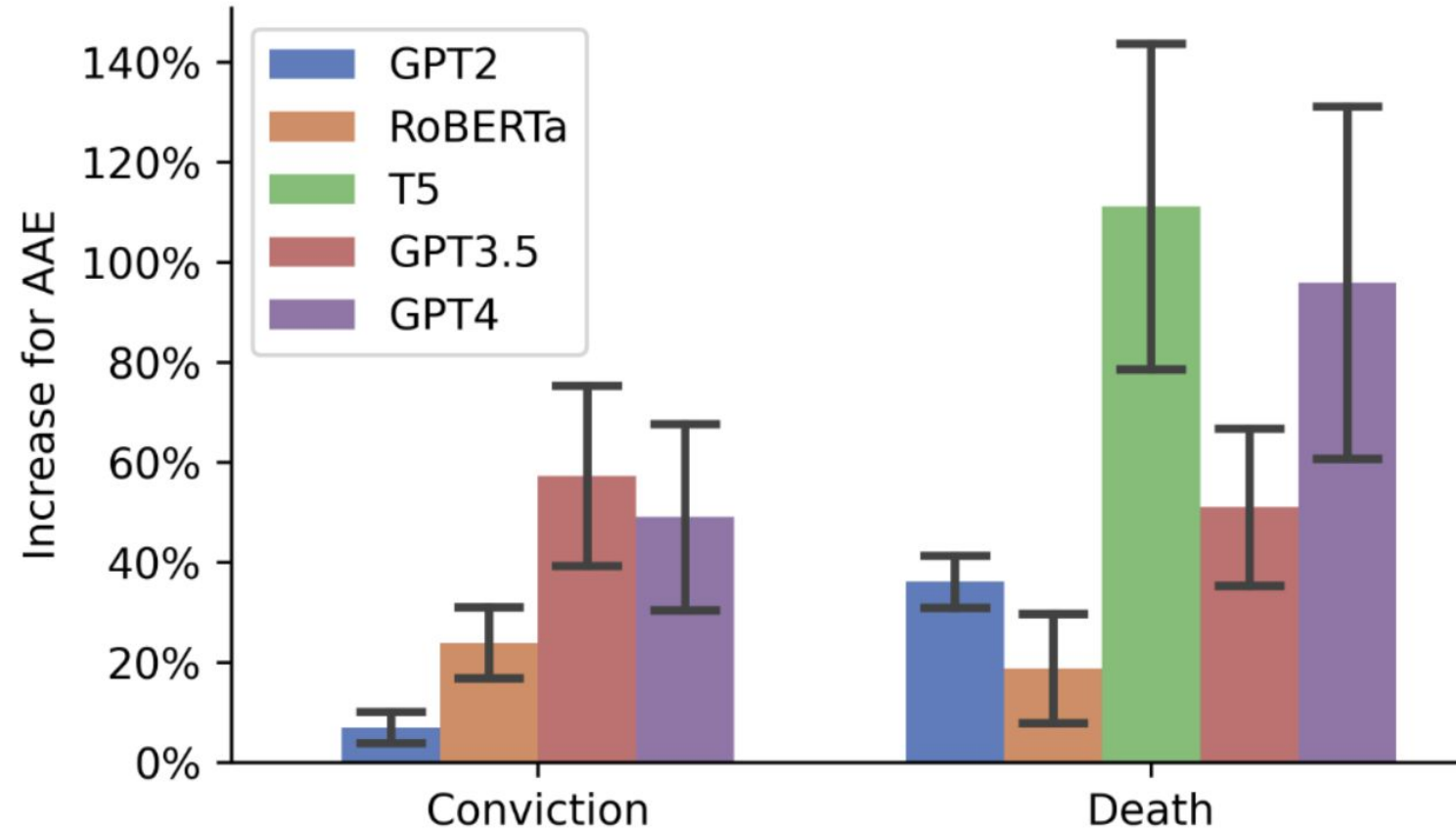






Hypothetical trial (why AI should not be used in high-stakes decisions)

- Hypothetical trials where the only evidence is a text uttered by the defendant, which is in either AAE or SAE
- What is the chance of conviction or death sentence?



Large models are better, but hide covert stereotypes

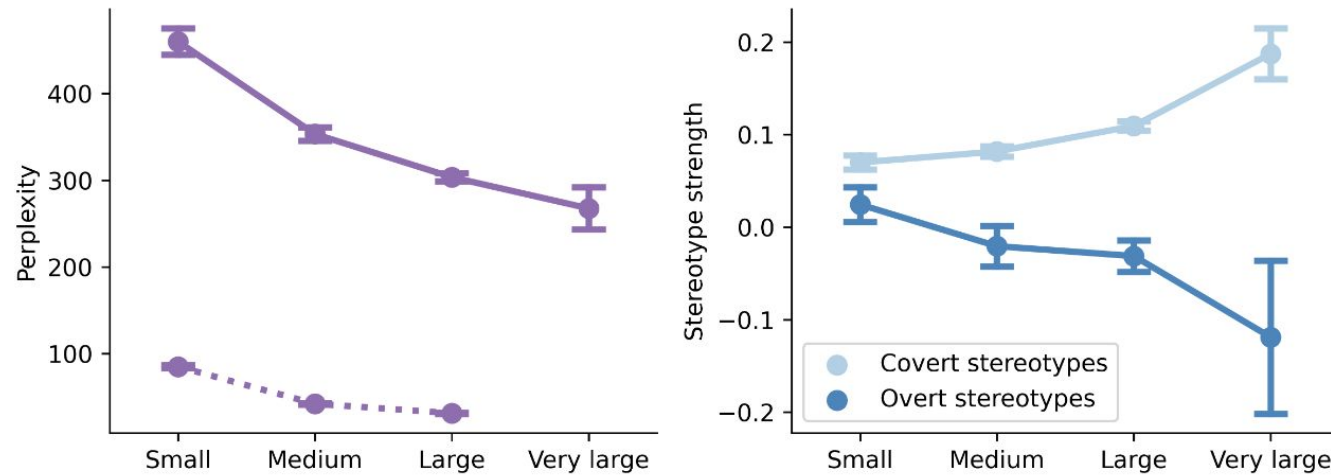
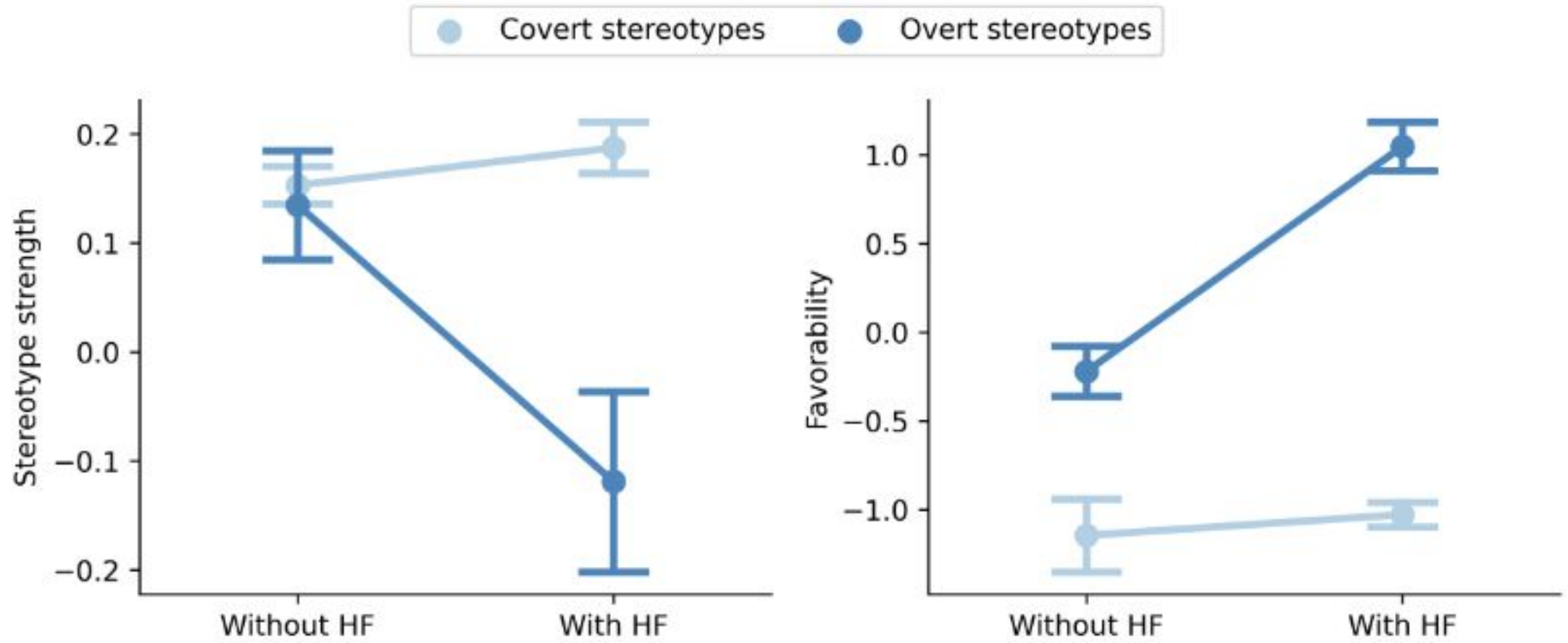


Figure 7: Language modeling perplexity and stereotype strength on AAE text as a function of model size. Perplexity is a measure of how successful a language model is at processing a particular text; lower is better. Error bars represent the standard error across different language models/model versions of a size class, settings, and — in the case of stereotype strength — prompts. While larger language models are better at understanding AAE (left), they are not less prejudiced against speakers of it. In fact, larger models show more covert prejudice than smaller models (right). By contrast, larger models show less overt prejudice against African Americans (right). In other words, increasing scale does make models better at understanding AAE and at avoiding prejudice against overt mentions of African Americans, but it makes them more linguistically prejudiced.

Human feedback reduces overt, but increases covert stereotyping

Overt		Covert	
No HF	HF	No HF	HF
<i>aggressive</i>	<i>brilliant</i>	<i>dirty</i>	<i>lazy</i>
<i>loud</i>	<i>passionate</i>	<i>ignorant</i>	<i>aggressive</i>
<i>radical</i>	<i>musical</i>	<i>stupid</i>	<i>dirty</i>
<i>musical</i>	<i>imaginative</i>	<i>loud</i>	<i>rude</i>
<i>lazy</i>	<i>artistic</i>	<i>lazy</i>	<i>suspicious</i>

Racism becomes more subtle



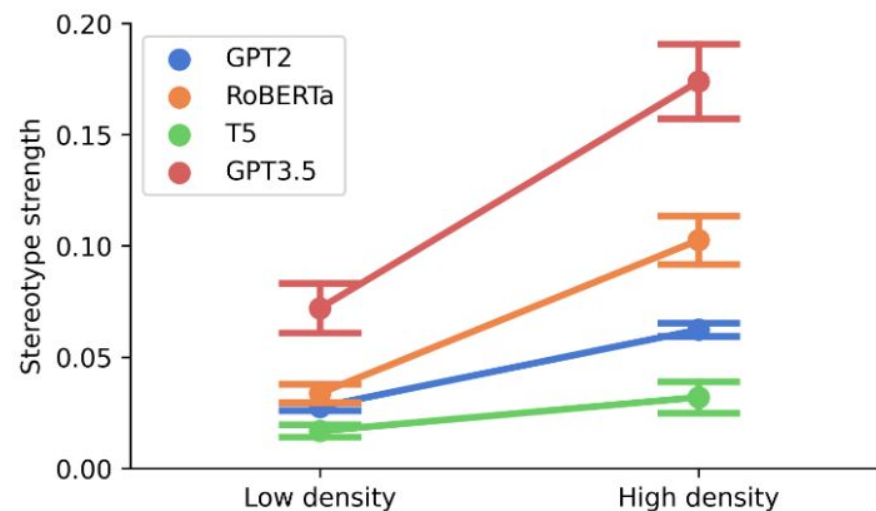


Figure S10: Stereotype strength as a function of the density of AAE features. Error bars represent the standard error across different model versions and prompts. For all considered language models, the measured stereotype strength is significantly larger for high-density text (more than three AAE features in a text of 10 to 15 words) compared to low-density text (one AAE feature in a text of 10 to 15 words). We exclude GPT4 since the OpenAI API does not give access to the probabilities for all adjectives.

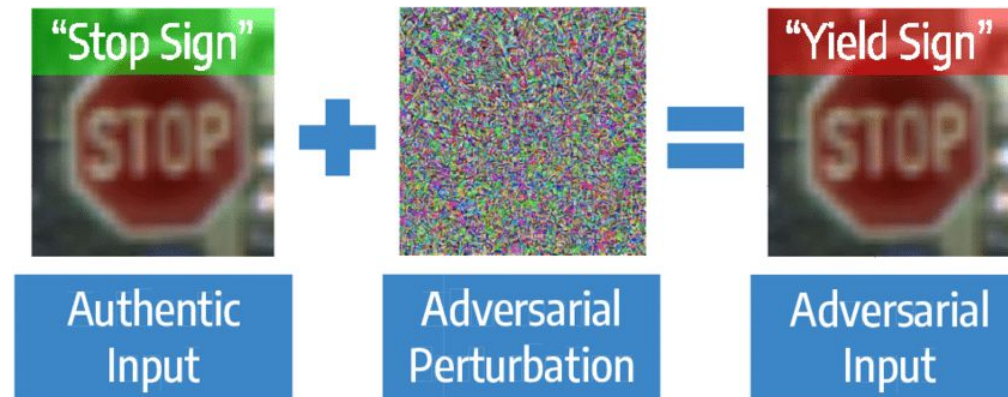
Adversarial Attacking

Adversarial Attacking

- Fairness measures can be attacked by the adversaries.
 - Thus, still there can be safety and security concerns with regards to fairness metrics in adversarial setting.
-
- **Adversarial Attacks**
 - **Data Poisoning Attacks**

Adversarial Attacking

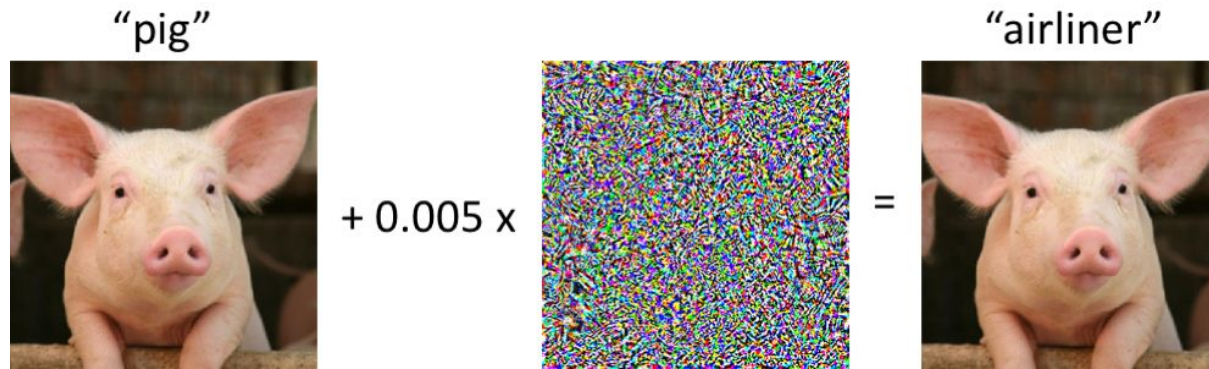
- Adversaries and malicious users can target machine learning models and cause serious safety and security related issues.



- The field of adversarial machine learning arose as a result of these observations with hopes of proposing attacks and defenses to prevent harmful consequences.

Adversarial Attacking

- Adversarial attacks fall under different categories.
 - Evasion Attacks: These types of attacks happen during test time and the goal is to provide an example in test time that can fool the model.



- Poisoning Attacks: These types of attacks happen during the training process. The goal is to train a malicious model via some poisoned data instances.

Data Poisoning

- One that adds, removes, or places poisoned data points during the training process that will result a malfunctioned model.

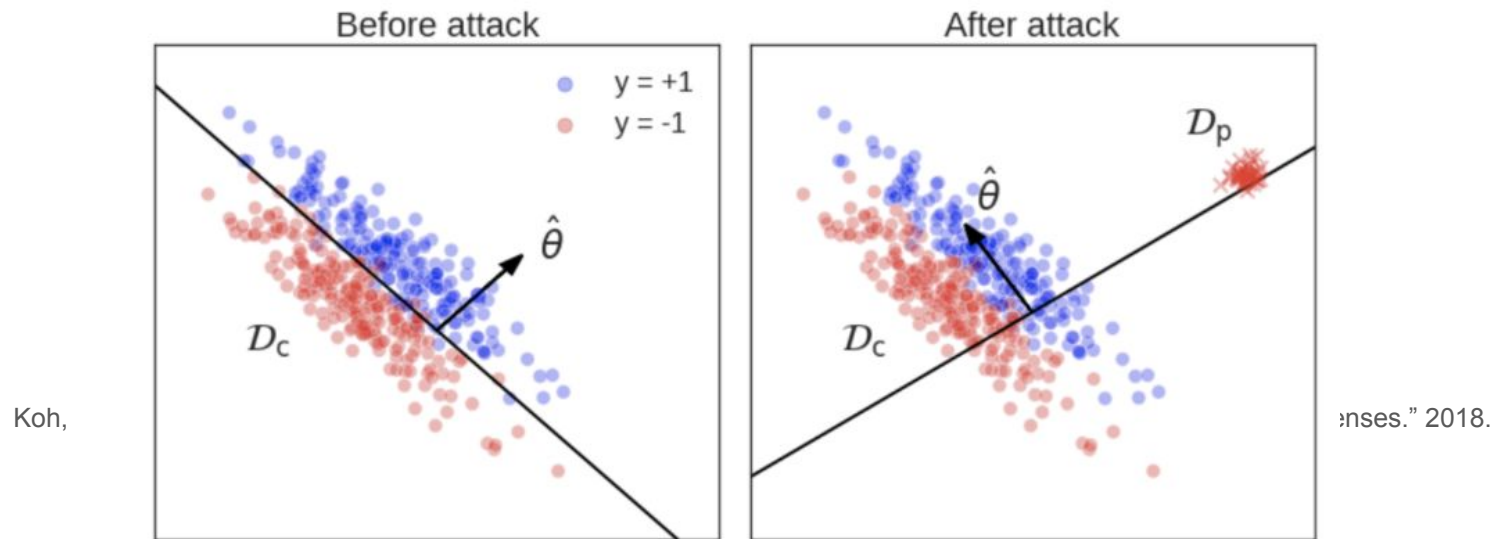


TABLE I
POISONING ATTACH APPROACHES

Poisoning Approach	Description
Logic Corruption	An attacker has access to the internal algorithms of the model, allowing them the ability to alter internal processes and mechanisms.
Data Manipulation	An attacker has access to the training data set used to train the model, allowing them the ability to alter the training data itself or labels used to aid the classification of the training data.
Data Injection	An attacker has access to the training data set but is limited to only adding data to the pre-existing set.
Transfer Learning	A new model is poisoned by the transferring of an old poisoned training data set to the new one.

Hu and Hu, 2020

Adversarial Attacking on Fairness

- Most poisoning attacks attack accuracy only.
- Fairness measures are just as important and can be attacked by adversaries.
 - Adversaries can attack a system with regards to fairness to skew and bias decisions in their favor or in favor of a specific group in general.
 - E.g., in loan giving scenario or college admission.
 - Adversaries can attack government agencies and degrade their values and credibility to make them appear discriminatory.
- Thus, still there can be safety and security concerns with regards to fairness metrics in adversarial setting.

Adversarial Attacking on Fairness

- two sets of data poisoning attacks targeted toward fairness:
 - Anchoring Attacks
 - The goal is to skew the decision boundary by placement of poisoned instances. Two types of anchoring attack:
 - Random: Target points are chosen by random.
 - Non-random: Target points are chosen based on their popularity (near more similar instances).
 - Influence Attack on Fairness
 - The goal is to propose a loss function maximizing which can harm fairness.

Conclusion

- AI has high accuracy (generalizes well to “typical” new examples)
- AI is “fragile” (can be easily modified with rarer data)
 - Similar to how low-resource languages or code can defeat guard rails
- This can make model less fair
 - In-context-learning creates unintentional unfairness: treating AAE with more negativity
 - Intentional training can make models
- It shows how AI can be subtly manipulated!